

Chapter 1: Websites

Today, the internet is accessible in almost every part of the world. In the last two decades, the internet and web have grown rapidly, so the websites. If you go two decades back, the websites were very different. They were not at all attractive, of course, and most importantly, they were static. By static, I mean, everything on a web page was fixed. But nowadays, websites are dynamic, generated by web applications.

Static and dynamic websites

In a static website, everything is fixed until someone changes it manually from behind. Such websites are created using HTML and are the most straightforward part of website development. All the users visiting a static website have the same view. **But the content on a dynamic website can be different for every user.** For example, amazon's homepage is a bit different for a signed-in user and a non-signed in user. If you are not signed in, you cannot see your account information, order history, and other stuff. It appears only when you sign in with your credentials.

A dynamic website is linked with at least one database where all the dynamic information is stored. There is no such database in the case of static websites. User-interaction is another essential part of a dynamic website.

HTML is used to create a static website.

HTML is one of the core technologies of the World Wide Web(WWW). The other two technologies are CSS and JavaScript. You can also use CSS and JavaScript on a static website to make it more attractive and a bit intractable. But the central concept, i.e. data is fixed and does not change. But using these three technologies more effectively, especially, JavaScript can create beautiful and high performing dynamic websites.

Don't worry; We will discuss all these three technologies in depth after this topic. But before moving further, let's talk HTML, CSS, and JavaScript in brief so you can have an idea of what you are going to learn.

HTML

HTML stands for Hypertext Markup Language. It does not matter how big or complicated your website is going to be; **you will always start with HTML**. It is the standard language to create structures for the web. While CSS and Javascript have changed a lot over the years, HTML of the 1990s and 2010s is not much different.

The basic structure of a web page is created using HTML. There are several HTML elements, and they are the building block of these pages. HTML elements are used in the form of tags. The tags are angular brackets with HTML names written inside them. For example, the HTML tag for image is `<img/>`. Most of these tags have a closing tag like `<p>` and `</p>`. However, some tags, such as `<img/>` does not require a closing tag. **CSS and JavaScript are further applied to HTML to change its appearance and to make it dynamic, respectively.**

CSS

Cascading Style Sheets or commonly known as **CSS is the presentation part of a web page**. HTML creates a structure, and CSS converts it into an attractive and more readable version. No website is complete without CSS today. Users expect a website to be appealing, engaging, and above all, properly readable.

With CSS, you can change the font, color, size, positions, layouts, and many more things. There are multiple ways of using CSS in an HTML file, each having its own advantage.

JavaScript

JavaScript is considered the most crucial part of a website. **It is the most popular language of the year 2019 according to StackOverflow insights**. Well, most of the websites you visit are created using javascript.

It is a scripting language that is used on client-side as well as server-side. Earlier, javascript can only run in a browser, but with the introduction of node.js, it can run outside too. Web frameworks and libraries such as Angular, React, Vue are built using javascript. As node.js, it is also used to create backend services.

Summary

- There are two types of websites - static and dynamic.
- Static websites have fixed content that does not change.

- Content in a dynamic website can change, either by users or automatically for different users.
- HTML, CSS, and JavaScript are the three core technologies of the World Wide Web(WWW).
- HTML elements are accessed using angular brackets, or commonly known as tags. These tags are used to create the structure for a web page.
- CSS is used to enhance the appearance of a web page.
- JavaScript is a scripting language that plays a vital role in developing dynamic websites. It is used for user interaction, content management, manipulating databases, and many more.

Chapter 2: Understanding HTML elements, tags, and attributes

As discussed in the last chapter, **HTML elements are the building blocks of a web page**. These elements are enclosed in angular brackets. Many HTML tags have corresponding closing tags. There are also a few tags that do not require such closing tags. We will discuss all these tags in the upcoming chapter, but first, you need to understand how HTML tags work.

Basic HTML tags

Let's start with the most basic tag, i.e. `<html>` tag. **Every HTML document starts with `<html>` tag and ends with its corresponding closing tag, `</html>` tag.** Other HTML tags are nested inside this tag only.

Other two basic HTML tags are `<head>` and `<body>` tags.

HTML files can render in a browser. The visible part in the browser window is written inside the `<body>` tag. It can contain several elements, such as paragraphs, headings, images, videos, sections, divisions, etc.

Another basic tag is the `<head>` tag. All the information regarding the document is listed in the `<head>` tag. It include HTML tags such as `<link>`, `<title>`, `<meta>`, `<style>`, etc. In the early versions, the `<head>` tag was mandatory but in HTML 5, It can be omitted.

```

1  <!DOCTYPE html>
2  <html>
3      <head>
4
5      </head>
6      <body>
7
8      </body>
9  </html>

```

This is how usually an HTML document is structured, the <head> tag first, followed by the <body> tag.

HTML attributes

All the HTML tags are built for a specific purpose. For example, the <p> is used to for paragraphs and is used for images. **Most of the HTML tags have additional properties or characteristics that are defined by attributes.** A tag may or may not have mandatory attributes. The tag, for example, must contain src and alt attributes. Further, you can place height and width attributes, but they are not mandatory. Have a look at the below HTML code.

```

1  <!DOCTYPE html>
2  <html>
3      <head>
4      </head>
5      <body>
6
7      <img src="" alt="" />
8
9      </body>
10 </html>

```

A tag is defined with two attributes - src and alt. Value for an attribute is written inside the double-quotes. As of now, these two attributes do not have any values.

Closing and opening tags

As I have mentioned above, Many tags have corresponding closing tags. **The difference between the opening and closing tags is that the closing tag has a forward slash.**

Some tags such as have a forward slash within itself only.

<!DOCTYPE html>

You can run HTML documents in a browser. The `<HTML>` tag defines that, it is an HTML document. But the browser needs to interpret the type of file. **The `<!DOCTYPE html>` is the declaration that informs the browser that it is an HTML document.**

`<!DOCTYPE html>` is not an HTML tag. You must declare it at the top of every HTML document. Also, to create an HTML file, you should save the file with the .html extension.

Summary

- The `<html>` tag is used to define an HTML document that contains all other tags.
- The content of an HTML document is defined inside the `<body>` tag.
- The `<head>` tag has all the information regarding the document.
- The attributes define the additional properties or characteristics for an HTML tag.
- The closing tag has a forward slash in it.
- Declare the `<!DOCTYPE html>` at the top and always save the file with .html extension.

Chapter 3: Paragraphs and headings

We can add a variety of content in an HTML document. **The most common content you can find on any web page is the simple text.** The text can be in any form or style. We can **create paragraphs of any length, headings of any size, and you can change color, font size, font style, background-color.**

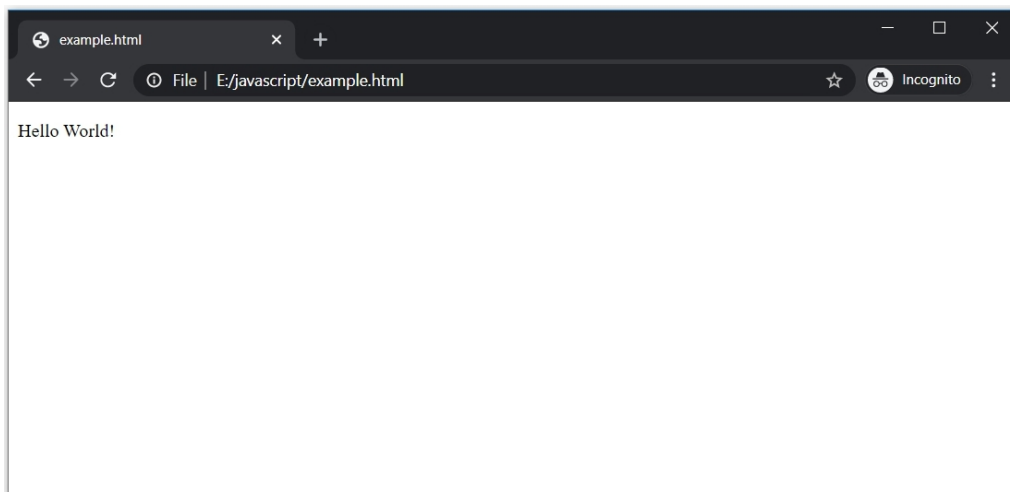
A paragraph in HTML is added using the `<p>` tag. For headings, we have multiple tags. These include `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>`, and `<h6>`, each of them having a different size.

`<p>` tag

A paragraph is a block of text that is created using the `<p>` tag.

```
1 <!DOCTYPE html>
2 <html>
3   <body>
4     <p>
5       | Hello World!
6     </p>
7   </body>
8 </html>
```

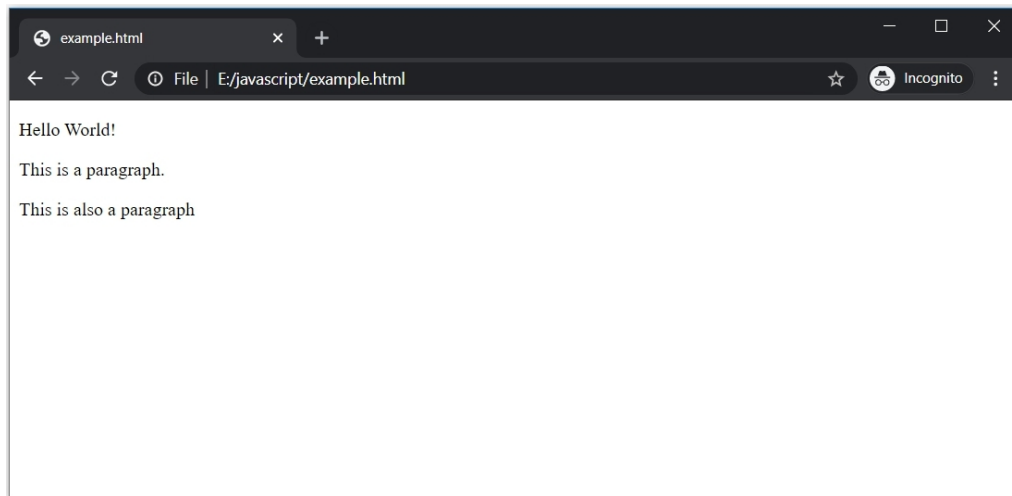
It has the corresponding closing tag on line 6 while the content is written between the tags. Let's check in the browser.



The content inside the tags is displayed on the browser. Let's add a couple of more paragraphs below it.

```
1 <!DOCTYPE html>
2 <html>
3   <body>
4     <p>
5       | Hello World!
6     </p>
7     <p>
8       | This is a paragraph.
9     </p>
10    <p>
11      | This is also a paragraph.
12    </p>
13  </body>
14 </html>
```

Now, there are three paragraphs in the document. Remember, each paragraph starts from a new line.



Line break

So each paragraph starts from a new line. But what if you want to add a new line inside a particular paragraph. Suppose we have the following text.

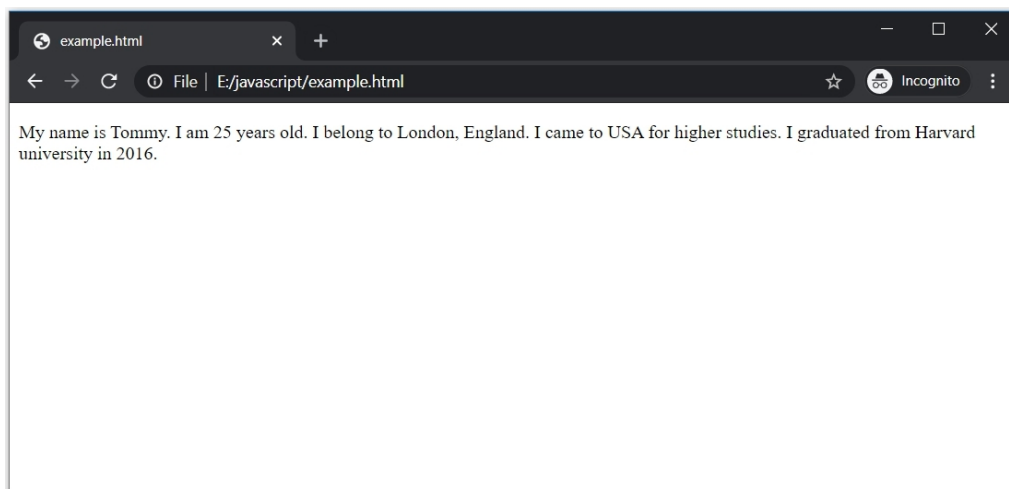
My name is Tommy. I am 25 years old. I belong to London, England.

I came to USA for higher studies.

I graduated from Harvard university in 2016.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <p>
6          My name is Tommy. I am 25 years old. I belong to London, England.
7          I came to USA for higher studies.
8          I graduated from Harvard university in 2016.
9      </p>
10 </body>
11
12 </html>
```

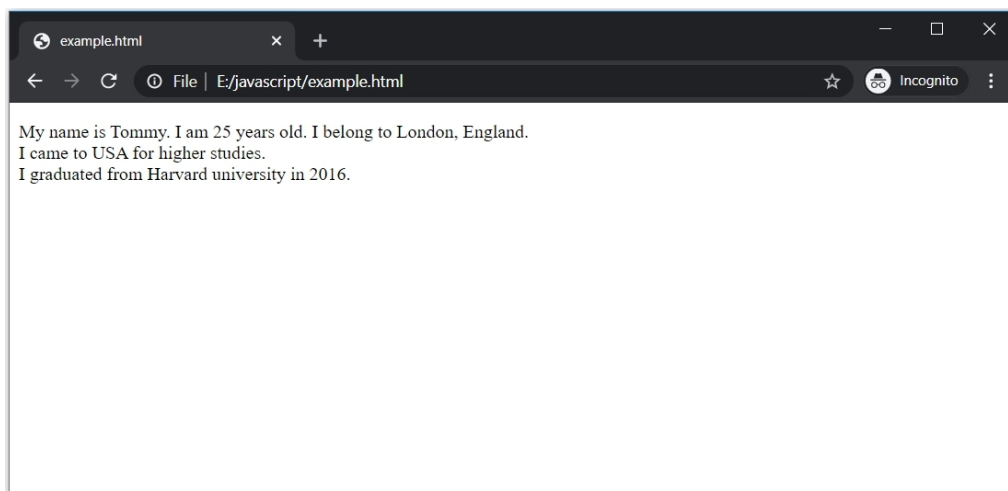
The lines 6,7, and 8 have new lines. Do you think it will display correctly in the browser?



It does not display the text in the same format as written in the `<p>` tag. Why? The reason is simple. It does not matter how we format the text in the `<p>` tag. It will always consider the whole content of a `<p>` tag as a single paragraph. **To add a new line, HTML provides the `<br>` tag.** Just place the tag at the end of the line where you want a new line to start.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <p>
6          My name is Tommy. I am 25 years old. I belong to London, England. <br>
7          I came to USA for higher studies. <br>
8          I graduated from Harvard university in 2016.
9      </p>
10 </body>
11
12 </html>
```

You can see, the `<br>` tag does not have a closing tag. It is an empty tag. It does not require a closing tag. Let's see what it displays in the browser.



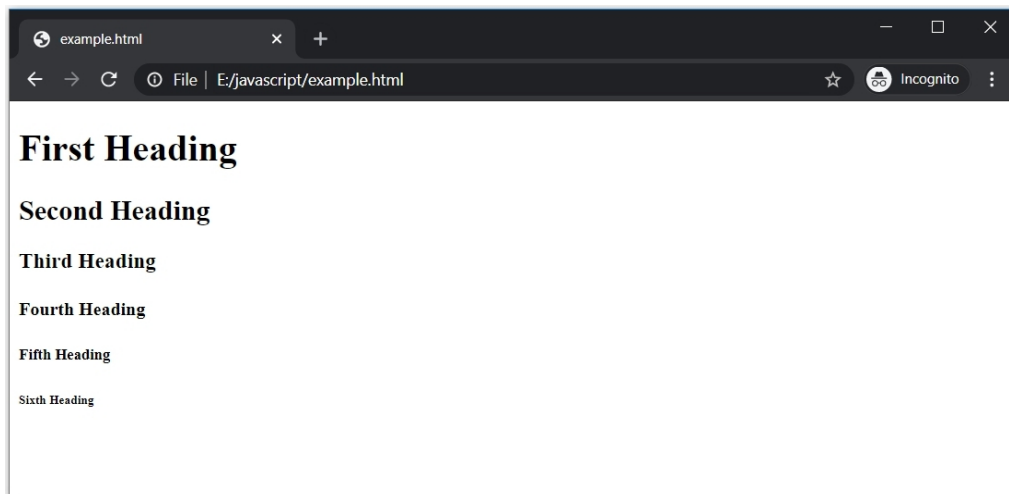
This is how I wanted the text.

`<h1>` and other heading tags

To give headings and subheadings, HTML provides the heading tags. They include `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>`, and `<h6>`. The size is the only difference between each of these tags.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <h1> First Heading</h1>
6      <h2> Second Heading</h2>
7      <h3> Third Heading</h3>
8      <h4> Fourth Heading</h4>
9      <h5> Fifth Heading</h5>
10     <h6> Sixth Heading</h6>
11 </body>
12
13 </html>
```

Lines 5 to 10 contain different types of headings.



The `<h1>` has the largest size while the `<h6>` has the smallest.

These tags are meant to provide headings and subheadings in a webpage. Headings are different from paragraphs. They are bold and big. **But that does not mean we should use them between paragraphs to make the text bold or big.** Use these tags efficiently.

Summary

- The `<p>` tags are used to write paragraphs.
- Each paragraph starts with a new line.
- To add a new line within a paragraph, use the `<br>` tag. This tag does not have any closing tag.
- There are six tags for headings. Each of them differs in size.
- Never use the heading tags between the paragraphs.

Chapter 4: Text formatting

While adding text in an HTML document, you may need to define special meaning for some parts. By special meaning, I mean, **pointing out a part of the text that appears different**.

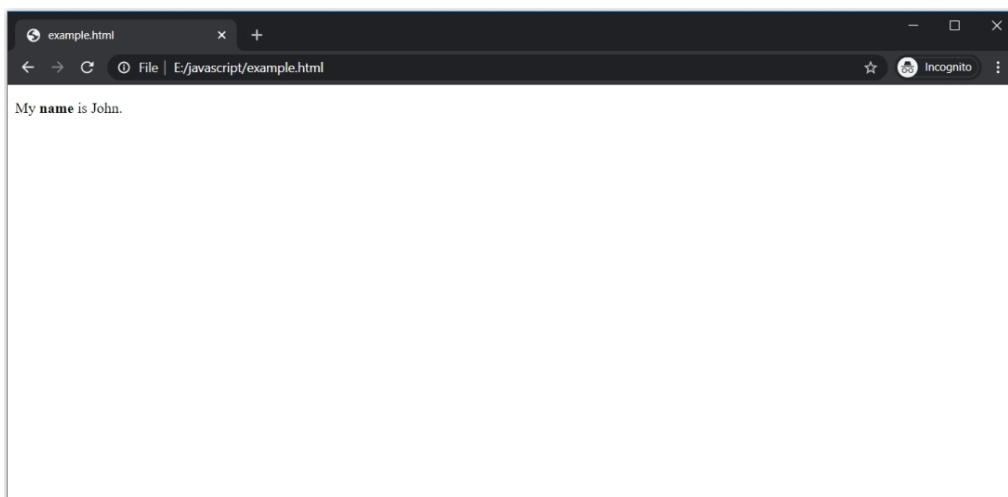
HTML provides several tags for formatting the text.

 and tags

The tag defines bold text.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5  |   <p>My <b> name </b> is John.</p>
6  </body>
7
8  </html>
```

In the paragraph, one word, i.e. 'name' is enclosed within the tag.



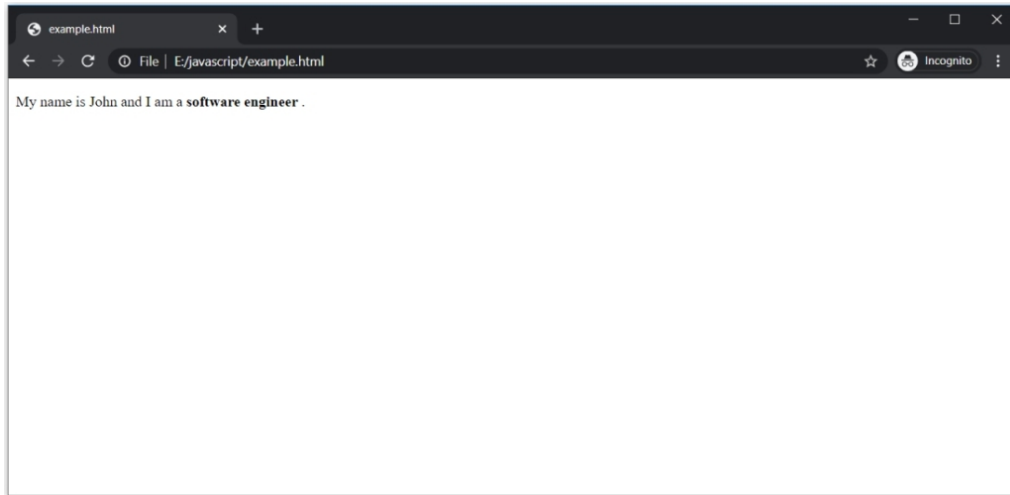
The text enclosed within the tag is bold now. Similarly, there is another tag that behaves in the same. It is called the tag. But, **the also defines that the text has extra importance**.

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <p>My name is John and I am a <strong> software engineer </strong> </p>
7
8  </body>
9
10 </html>

```

Let's see what happens when the text is enclosed within the `<strong>` tag.



You may not find any difference between `<b>` and `<strong>`, but the strong text has an extra meaning while the bold don't.

`<i>` and `<em>` tags

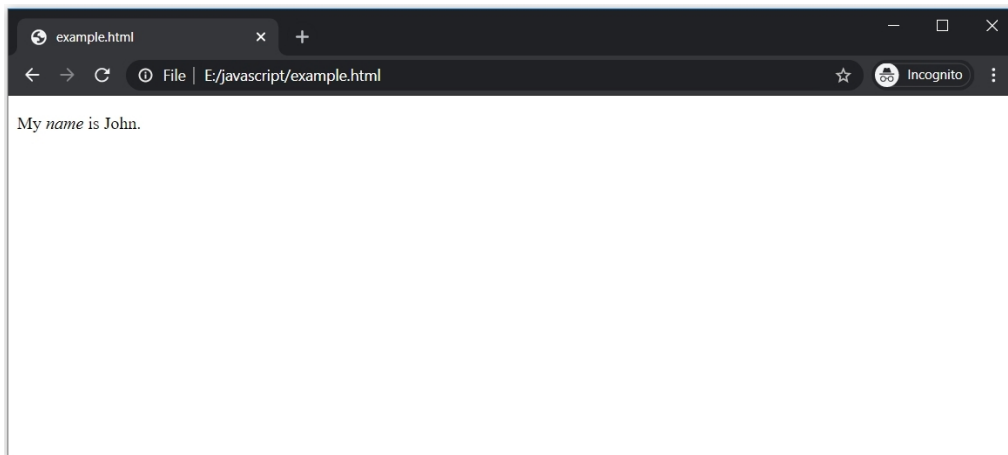
The `<i>` tag defines italic text.

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <p>My <i> name </i> is John </p>
7
8  </body>

```

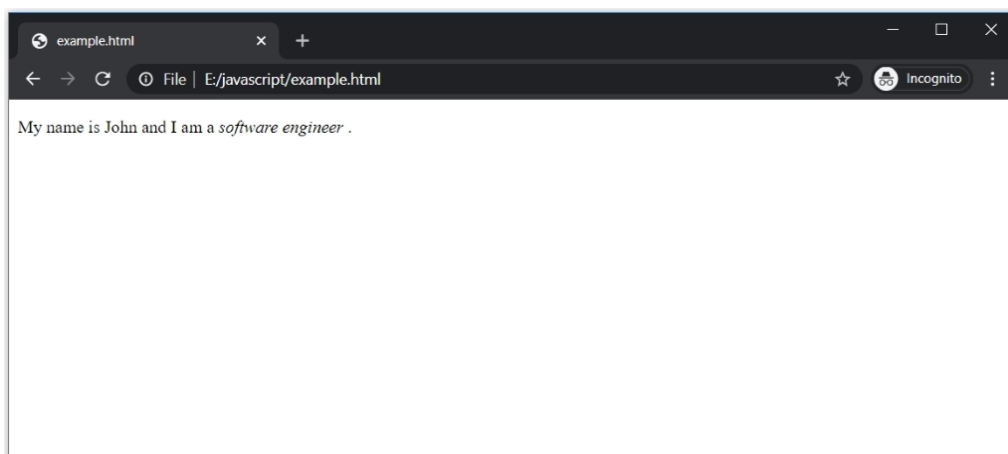
In the paragraph, one word, i.e. 'name' is enclosed within the `<i>` tag.



Similarly, there is another tag, ****, which also define italic text, but with extra importance.

```
1 <!DOCTYPE html>
2 <html>
3
4 <body>
5
6     <p>My name is John and I am a <em> software engineer </em>.</p>
7
8 </body>
```

Let's see what happens when the text is enclosed within the **** tag.



There does not appear any difference, but the text within the **** tag has extra importance, similar to the **** tag.

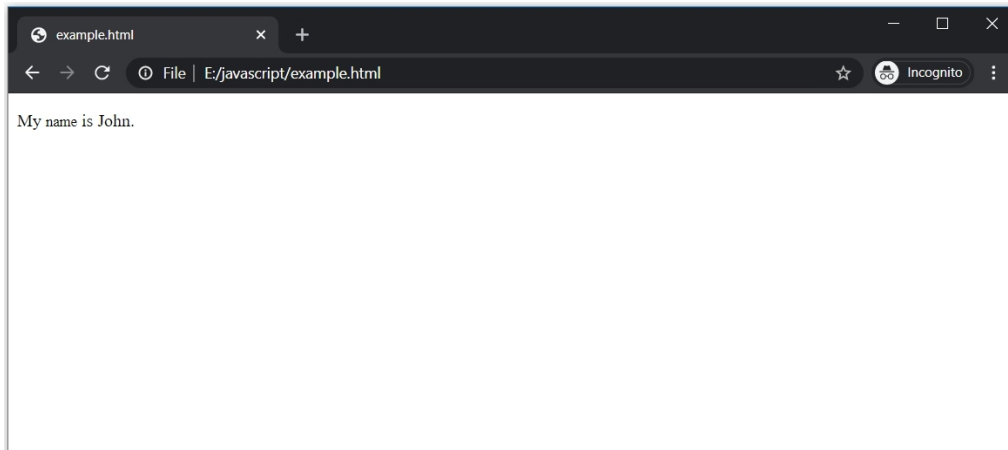
<small> tag

Sometimes, you may need to define a text in a small size when compared to other text. **The <small> tag in HTML define small text.**

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <p>My <small>name</small> is John.</p>
7
8  </body>
9
10 </html>

```



You can see, the word - 'name', appears smaller than the rest of the text.

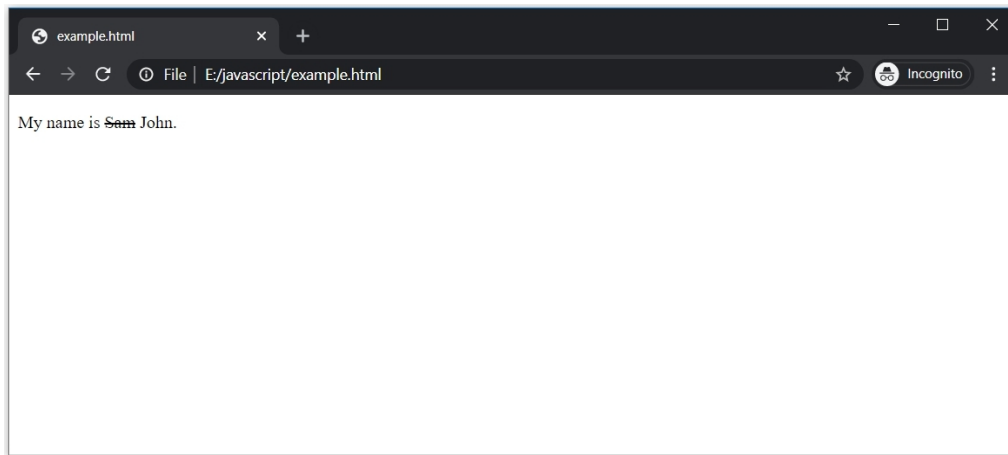
 tag

Did you ever cut a word or sentence while writing? Similarly, **HTML provides the tag to present a deleted or removed text.**

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <p>My name is <del>Sam</del> John.</p>
7
8  </body>
9
10 </html>

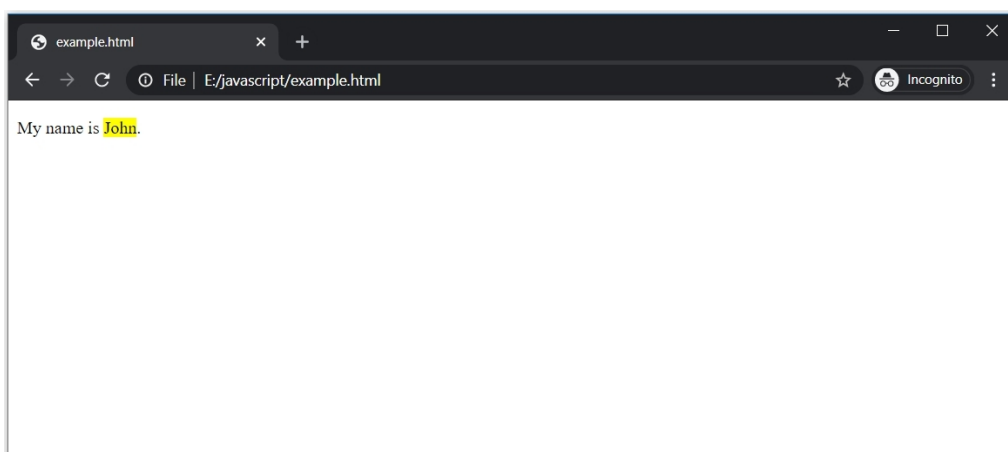
```



<mark> tag

Highlighting is one of the most common ways of pointing out a subtext from a text. In HTML, **highlighting or marking text can be achieved by using the <mark> tag.**

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <p>My name is <mark>John</mark> </p>
7
8  </body>
9
10 </html>
```

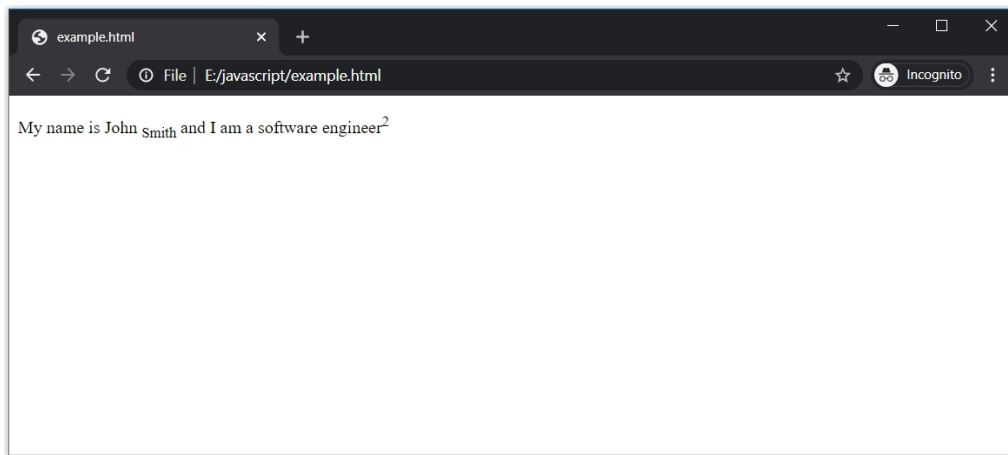


The highlighted text is visible in yellow.

<sub> and <sup> tags

The `<sub>` tag defines subscripted text while the `<sup>` tag defines superscripted text.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <p>My name is John <sub>Smith</sub> and I am a software engineer<sup>2</sup></p>
7
8  </body>
9
10 </html>
```



Summary

- The `<b>` and `<strong>` tag defines bold text but the `<strong>` tag also means extra importance.
- The `<i>` and `<em>` tag defines italic text but the `<em>` tag also means extra importance.
- The `<small>` tag defines a smaller text in comparison with other text.
- The `<del>` tag is used to represent a deleted or removed text.
- The `<mark>` tag defines highlighted or marked text in yellow.
- The `<sub>` and `<sup>` tags defines subscripted and superscripted text respectively.

Chapter 5: Hyperlinks

There are multiple web pages on a website, right? We can navigate from one page to another. **The links in HTML allow a user to navigate from one web page to another. Such links in HTML are called hyperlinks.**

The <a> is used to create hyperlinks in HTML. In this chapter, we will discuss how to use the <a> tag to move from one HTML document or web page to another.

<a> tag

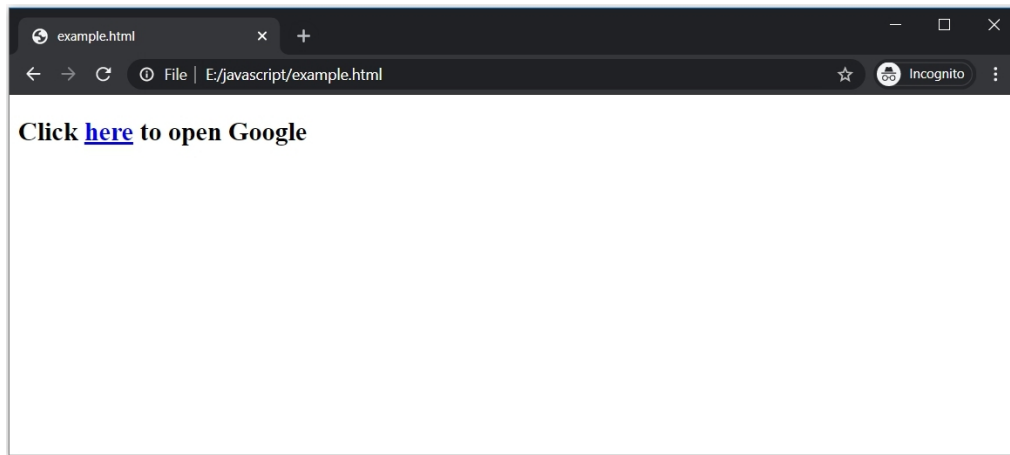
The <a> tag has few attributes. **One of these attributes - href, is a mandatory attribute that holds the link of the document or web page that will open when clicked.** Observe the syntax of the <a> tag.

```
3   <a href="<url>"> text </a>
```

The URL should be written within the quotes. The text will appear on the screen and when clicked on it, the URL specified for the href attribute will open. Observe the following HTML code.

```
1   <!DOCTYPE html>
2   <html>
3
4   <body>
5
6       <h2> Click <a href="https://google.com"> here </a> to open Google </h2>
7
8   </body>
9
10  </html>
```

In the paragraph, the word - 'here' is a hyperlink. The href is 'https://google.com'. This means the homepage of Google will open when clicked on the hyperlink.



Did you notice something different with the appearance of the hyperlink? **It is underlined and blue.** There is a way to remove this styling by using CSS. We will discuss it later in the CSS section.

You can also move locally from one HTML document to another. In the href attribute, you have to provide the proper path for the document you want to navigate.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <h2> Click <a href="demo.html"> here </a> to open another document </h2>
7
8  </body>
9
10 </html>
```

In the above <a> tag, the name of a document is specified that happens to be in the same folder. Clicking on the hyperlink will open the demo.html document.

target attribute

The target attribute is an optional attribute of the <a> tag. It specifies where to open the document or web page. It can have the following four values:

- **_self** : To open the document or web page in the same tab/window. The target is set to _self by default.
- **_blank** : To open the document or web page in a new tab/window.
- **_top** : To open the document or web page in the full body of the window.
- **_parent** : To open the document or web page in the parent frame.

You can also provide a framename as the value of the target attribute to open the document in a particular frame.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <h2> Click <a href="https://google.com" target="_blank"> here </a> to open Google </h2>
7
8  </body>
9
10 </html>
```

The hyperlink in the above paragraph will **open in a new tab/window**.

title attribute

The title attribute provides a title for a hyperlink. Whenever the mouse hovers over the hyperlink, it will display a text which is called the title. By default, it does not show anything.

The title is usually extra information about the hyperlink in a very short form.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <h2> Click <a href="https://google.com" title="Go to Google"> here </a> to open Google </h2>
7
8  </body>
9
10 </html>
```

More with <a> tag

The <a> tag is not limited to text. You can also use images and buttons as hyperlinks. Even, you can use a part of an image as a hyperlink.

Summary

- The <a> tag in HTML is used to create hyperlinks.
- The href attribute is mandatory because it holds the URL of the document or web page that will open.
- The target attribute specifies where the document or web page will open.
- Use '_blank' as the value of the target attribute to open the document in a new tab/window.
- The title holds the extra information that will appear when hovered over the hyperlink.
- Images and buttons can also work as hyperlinks.

Chapter 6: Images

Images are the most common part of a website after the text. Today, almost every website you visit has images. Images improve the appearance of a website and attract more audiences.

To simply add an image, **use the tag**. But, it is not always just adding an image. The image should have proper alignment, position, and size.

 tag

The tag has the following syntax.

```

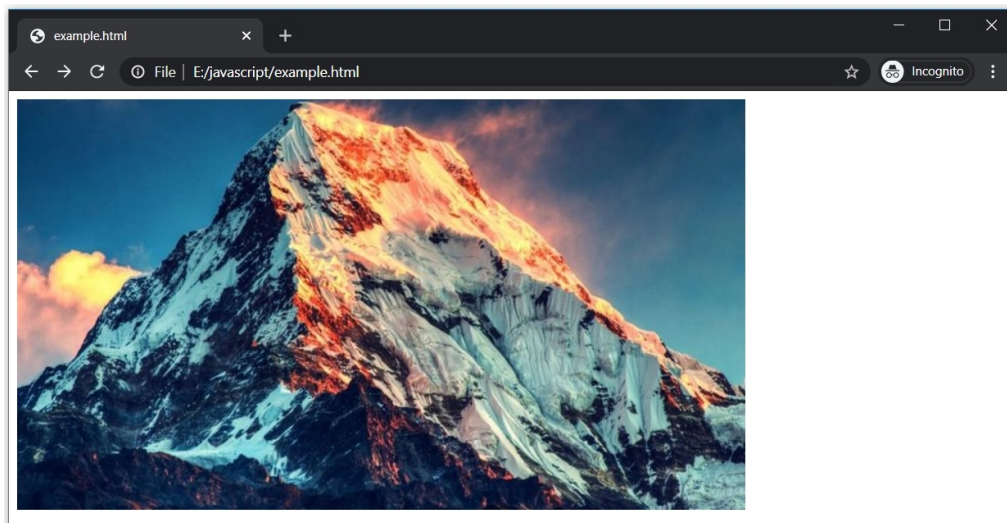
```

The tag does not have a corresponding closing.

The src attribute is mandatory because it holds the URL of the image. The image can be **present locally or it can be another server**. Let's discuss how to add local images in an HTML document.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      
7
8  </body>
9
10 </html>
```

The image is present in the 'images' folder. This folder is in the same location as the HTML file. It needs to be a proper path or the image won't appear on the screen. Moreover, **the extension of the image should also be present after the name**.



Similarly, we can add images from another server.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      
7
8  </body>
9
10 </html>
```

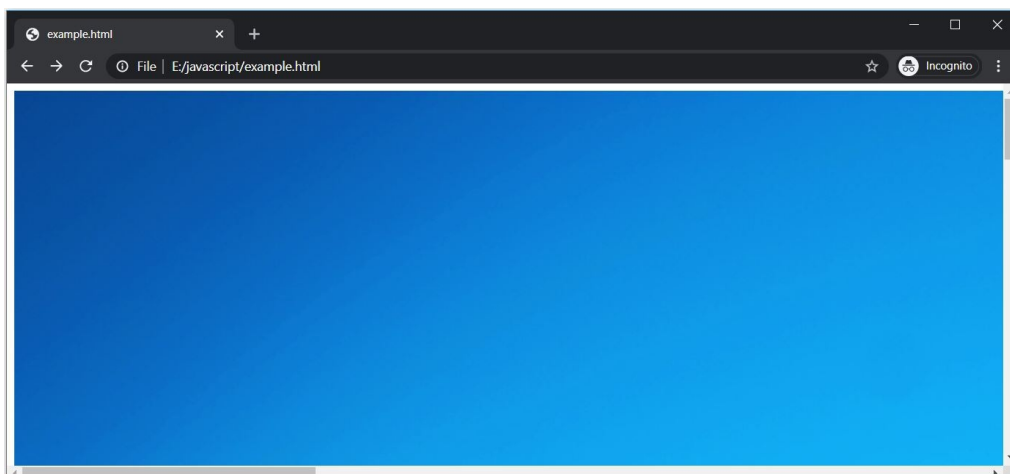
Just remember, the link should be working.

Image size

Have a look at the following image.



Let's see how this image appears in the browser.



This is not how it should appear, right? So what do you think is wrong here? By default, the size of the image appearing on the screen is the same as its actual size. This image is huge in size and that is why only a portion of it is appearing on the screen. The whole image can be seen by scrolling up and down.

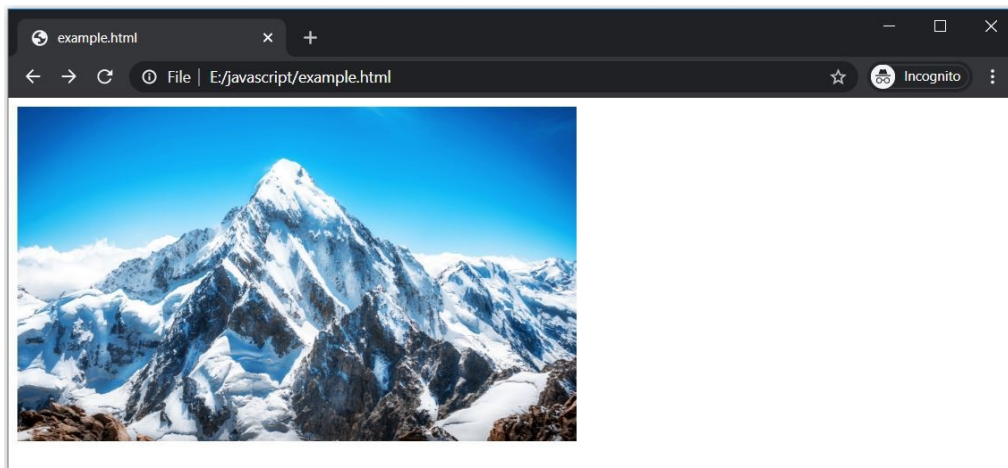
To encounter such situations, **the `<img>` tag has two attributes to set the width and height of an image. They are simply called width and height.**

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      
10
11 </body>
12
13 </html>

```

The height and width are set to 300 and 500, respectively.



Now, it looks better.

Similarly, we can use CSS for image styling which we will discuss in the CSS section.

alt attribute

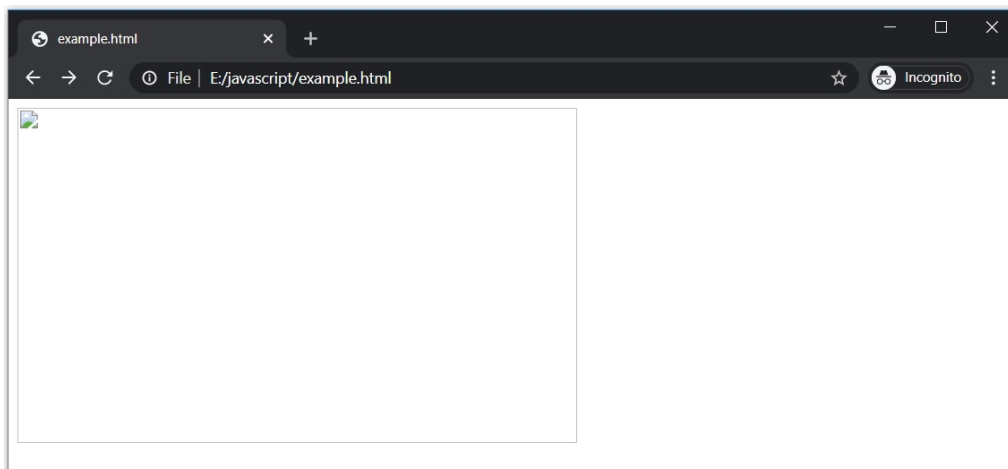
The alt attribute is optional but it is recommended to use alt in every tag. Suppose, an image fails to load for some reason. Then what? Let's see.

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      
10
11 </body>
12
13 </html>

```

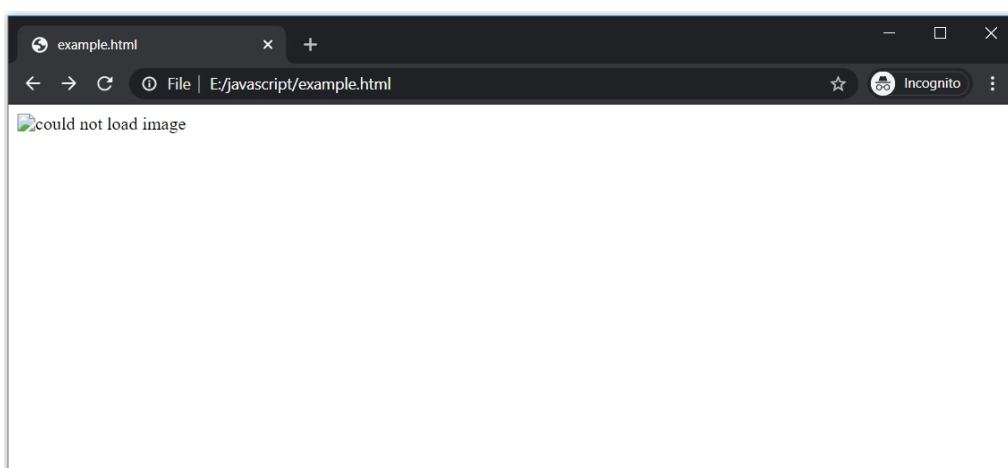
I added a few characters in the URL of the image to make it invalid.



This does not look nice, right?

So when an image fails to load, **the value of the alt attribute appears in its place.**

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      
11
12 </body>
13
14 </html>
```



Now, the user will know if there is a problem with image loading.

Image as links

As discussed in the previous chapter, images can also work as links. **Simply, add image inside the <a> tag.** The rest of the procedure is the same.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <a href="https://google.com">
7          
11      </a>
12
13 </body>
14
15 </html>
```

Summary

- Images in HTML are added using the tag.
- The src attribute holds the URL or path of the image.
- To change the size of an image, use the height and width attributes.
- The alt attribute holds the value that will be displayed if the image fails to load.
- Images can also be used as links in HTML.

Chapter 7: Tables

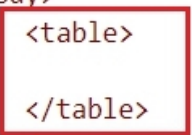
Data on a web page should be represented in a proper manner. There are various ways of representing data. We already discussed some of these ways, such as the `<p>` and `<h1>` tags. But sometimes, we may need a better way to display data more accurately and clearly. **Tables are considered one of the best ways to represent data.**

Tables arrange data in rows and columns. In HTML, **the `<table>` tag is used to create a table.** There are few other tags and attributes that are necessary to create a proper table in HTML. So, in this chapter we will discuss how to create a simple table in HTML.

The `<table>` tag

The `<table>` tag alone does not display anything.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <table>
6
7      </table>
8  </body>
9
10 </html>
```



It only defines a table. We need to create rows and columns manually.

HTML has the `<tr>` tag and `<td>` tag to create a row and column, respectively.

The `<tr>` and `<td>` tags

As mentioned, the `<tr>` tag creates a row. It is placed inside the `<table>` tag.

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <table>
6          <tr>
7              </tr>
8          </tr>
9      </table>
10
11 </body>
12

```

The above table has one row. Let's add two columns in this row using the `<td>` tag.

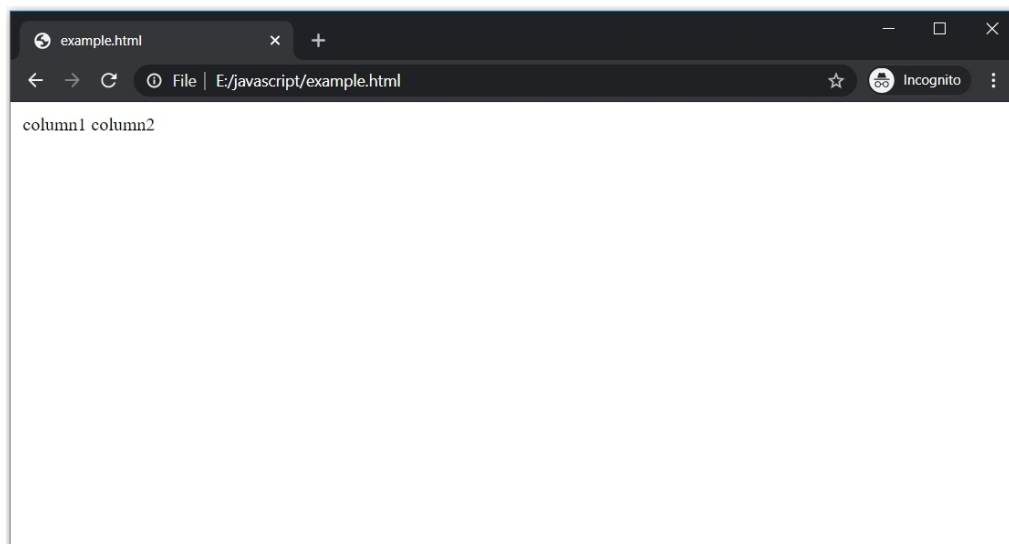
```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <table>
6          <tr>
7              <td>
8                  </td>
9              <td>
10                 </td>
11            </tr>
12        </table>
13
14 </body>
15
16 </html>

```

So, this is how rows and columns are created in HTML. Let's add some content in each column and see how the tables appear in the browser.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <table>
6
7          <tr>
8              <td>
9                  column1
10             </td>
11             <td>
12                 column2
13             </td>
14         </tr>
15     </table>
16
17 </body>
18
19
20 </html>
```



This does not look like a table, right? By default, a table in HTML does not have any border. **The CSS is required to add a border to the table.**

Table border

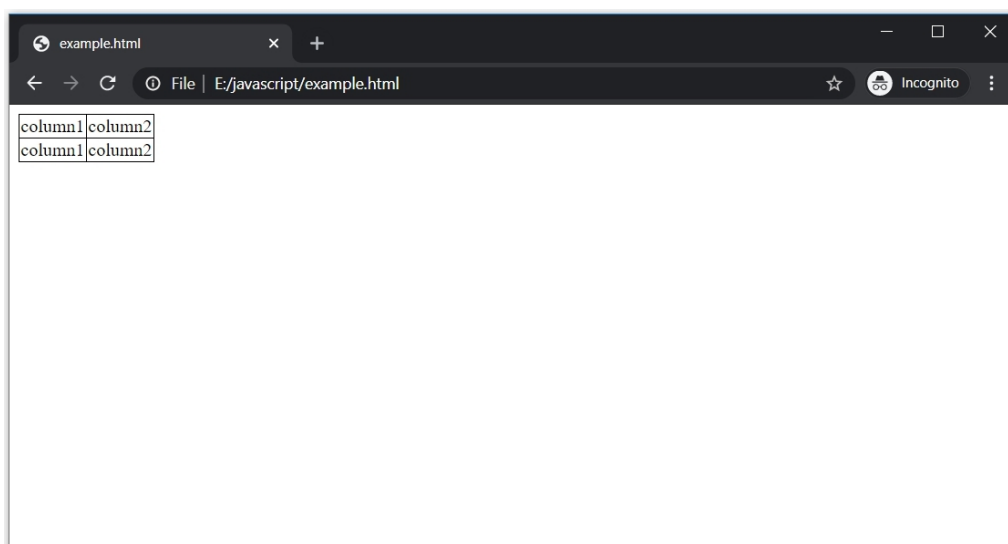
We haven't discussed CSS yet. But we will add borders in the table by using some CSS.

```

1  <!DOCTYPE html>
2  <html>
3  <head>
4      <style>
5          table, th, td {
6              border: 1px solid black;
7              border-collapse: collapse;
8          }
9      </style>
10 </head>
11
12 <body>
13     <table>
14         <tr>
15             <td>
16                 column1
17             </td>
18             <td>
19                 column2
20             </td>
21         </tr>
22     </table>
23 </body>
24
25 </html>

```

In the `<head>` tag, a `<style>` tag is added and it has some CSS in it. Ignore this part for now. We will discuss it later in the CSS section. For now, observe the table in the browser.



Now it looks like a table. A second row is also added.

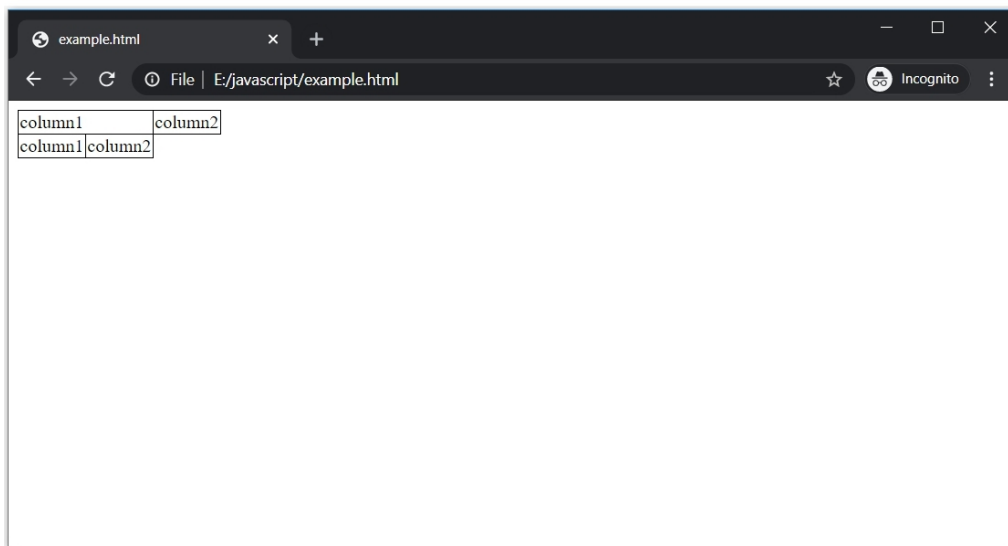
colspan and rowspan attributes

Tables depend a lot on CSS. We can create attractive and better tables with CSS but HTML also provides some nice attributes that could be used with tables. One of these attributes is the colspan attribute.

The colspan attribute increases the span of a cell according to the value specified.

```
14      <table>
15
16      <tr>
17          <td colspan="2">
18              column1
19          </td>
20          <td>
21              column2
22          </td>
23
24      </tr>
25
26      <tr>
27          <td>
28              column1
29          </td>
30          <td>
31              column2
32          </td>
33
34      </tr>
35
36  </table>
```

The first cell of the first row has colspan of 2.



You can see that the cell now covers the area of two columns.

The `colspan` attribute can have any value but it should be used carefully or the result can be an unpleasant table.

Similarly, **the `rowspan` is used to increase the span of a row.**

The `<th>` tag

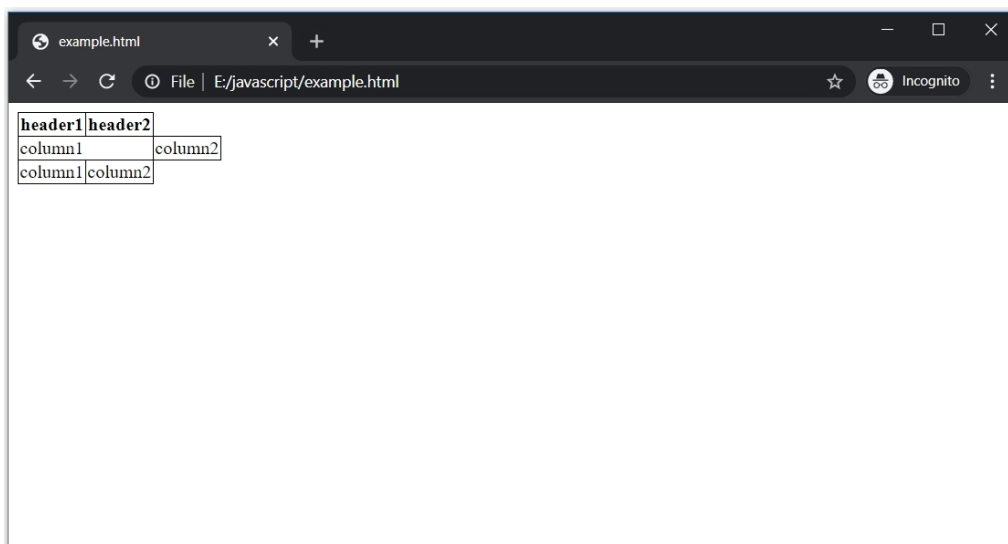
The `<th>` tag is used to create a header . It behaves like the `<td>` tag but the content is bold.

```

13      <table>
14      |   <tr>
15      |   |   <th>
16      |   |   |   header1
17      |   |   </th>
18      |   |
19      |   |   <th>
20      |   |   |   header2
21      |   |   </th>
22      |   |
23      |   </tr>
24      |   <tr>
25      |   |   <td colspan="2">
26      |   |   |   column1
27      |   |   </td>
28      |   |   <td>
29      |   |   |   column2
30      |   |   </td>
31      |   </tr>
32      |   <tr>
33      |   |   <td>
34      |   |   |   column1
35      |   |   </td>
36      |   |   <td>
37      |   |   |   column2
38      |   |   </td>
39      |   </tr>
40      </table>

```

Now, the first row has headers instead of normal cells.



You can see, the text is bold in the headers.

The `<thead>` and `<tbody>`

The `<thead>` and `<tbody>` tags are used to group the head part and the body of the table.

The table created has three rows - first one has headers while the other two have columns. So the first one can be written inside the `<thead>` tag and others can be written inside the `<tbody>` tag.

```
13  <table>
14      <thead>
15          <tr>
16              <th>
17                  header1
18              </th>
19
20              <th>
21                  header2
22              </th>
23          </tr>
24      </thead>
25
26      <tbody>
27          <tr>
28              <td colspan="2">
29                  column1
30              </td>
31              <td>
32                  column2
33              </td>
34          </tr>
35          <tr>
36              <td>
37                  column1
38              </td>
39              <td>
40                  column2
41              </td>
42          </tr>
43      </tbody>
44  </table>
```

These tags won't affect the table but is a good practice to group the rows of a table in HTML.

Summary

- Tables are used to arrange data in rows and columns.
- HTML has the `<table>` tag to create tables.
- The `<tr>` tag creates a row while the `<td>` tag creates a cell or column.
- The `<th>` tag can also be used in the place of the `<td>` tag. It specifies a header cell with bold content.
- The `colspan` and `rowspan` attributes are used to increase the span of a cell or row, respectively.
- The `<thead>` and `<tbody>` tags are used to group header rows and body rows in tables.

Chapter 8: Lists

Another way of representing data is by using lists.

Lists are used commonly on web pages. Mostly, **they are used to represent related data.** There are three types of lists in HTML - **ordered, unordered, and description.**

Ordered list

An ordered list is always in some kind of order. **By default, an ordered list is numerical.** But It also has other options.

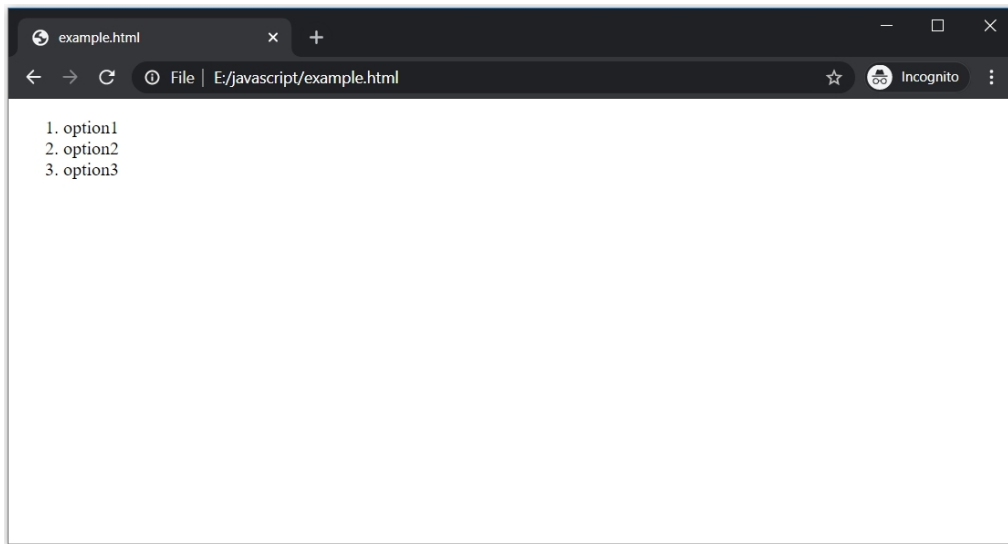
The ordered list is created using the `<ol>` tag.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <ol>
6
7      </ol>
8
9  </body>
10
11 </html>
```

The `<ol>` tag only defines an ordered list. To insert values, we have to use the `<li>` tag.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <ol>
6
7          <li> option1 </li>
8          <li> option2 </li>
9          <li> option3 </li>
10
11      </ol>
12
13 </body>
14
15 </html>
```

Now, the list has three values. Remember, each value should be written inside the `<li>` tag.

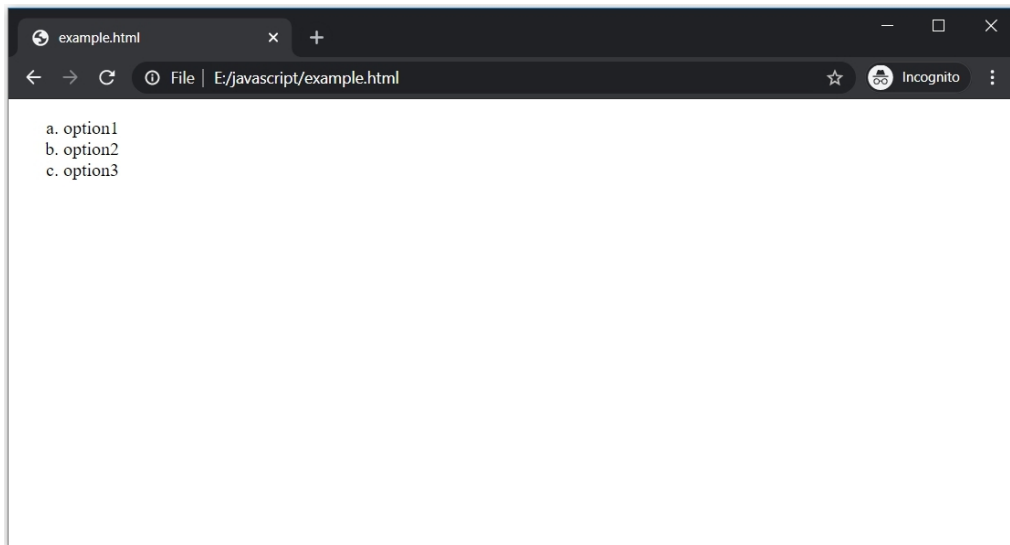


Type attribute in the ordered list

Every option starts with a number in ordered list. **We can change this to alphabets and roman numbers - both lowercase and uppercase.** To do this, use the type attribute.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <ol type="a">
6
7          <li> option1 </li>
8          <li> option2 </li>
9          <li> option3 </li>
10
11      </ol>
12
13 </body>
14
15 </html>
```

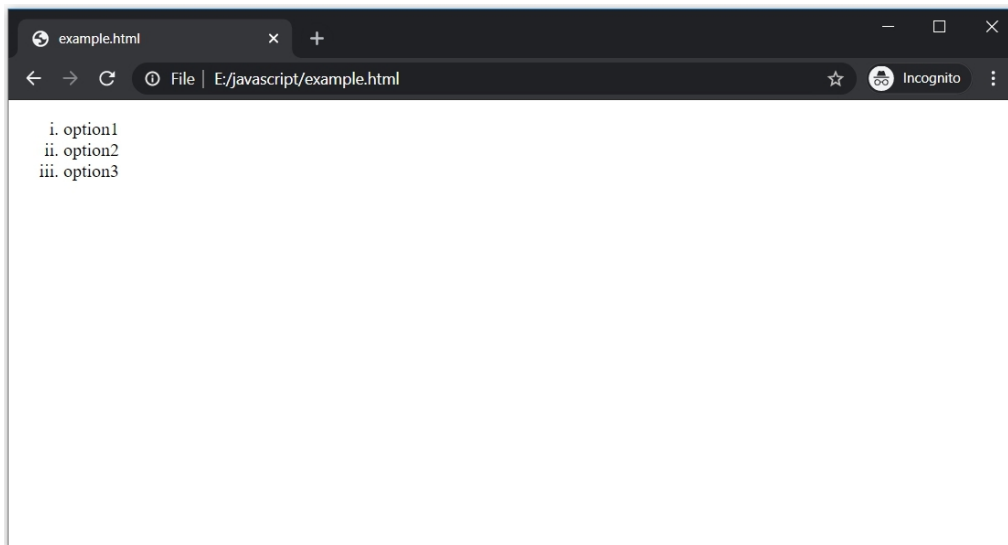
The type attribute has 'a' as its value.



The numbers are replaced by lowercase alphabets.

Similarly, use 'i' for lowercase roman numbers.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <ol type="i">
6
7          <li> option1 </li>
8          <li> option2 </li>
9          <li> option3 </li>
10
11      </ol>
12
13 </body>
14
15 </html>
```



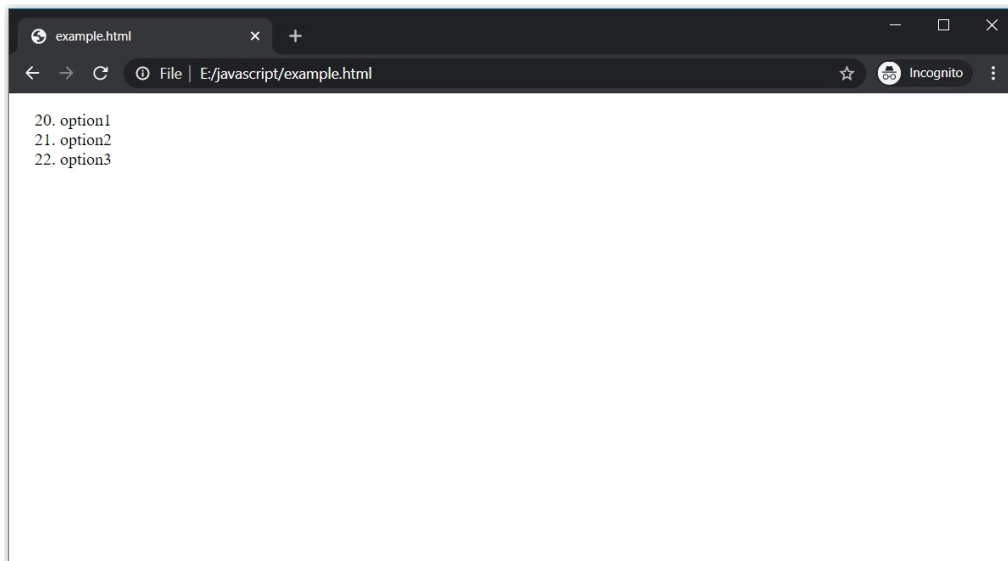
Other options are; 'A' for uppercase alphabets and 'T' for uppercase roman numbers.

start attribute

A numerical list always starts from 1. Similarly, an alphabetic list starts from 'a' or 'A'. **We can also control the starting point of an ordered list by using the start attribute.**

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <ol start = "20">
6
7          <li> option1 </li>
8          <li> option2 </li>
9          <li> option3 </li>
10
11      </ol>
12
13 </body>
14
15 </html>
```

The value of the start attribute is 20.



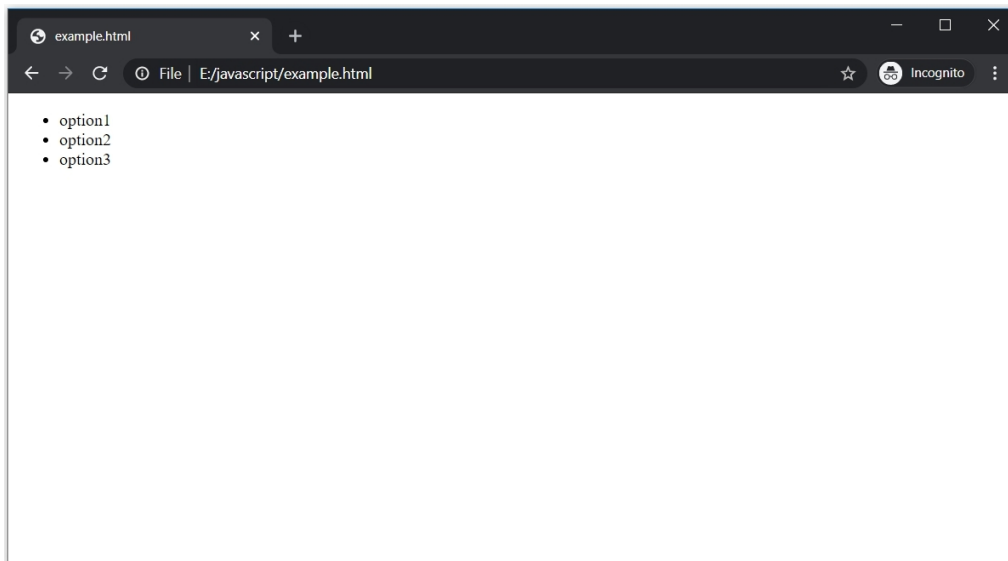
The list starts with 20. Similarly, we can use the start attribute for any type.

Unordered list

An unordered list does not have any order and it is a bulleted list.

The `<ul>` tag is defined using the `<ul>` tag and `<li>` tag is used for options just like the ordered list.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <ul>
6          <li> option1 </li>
7          <li> option2 </li>
8          <li> option3 </li>
9      </ul>
10
11
12
13 </body>
14
15 </html>
```



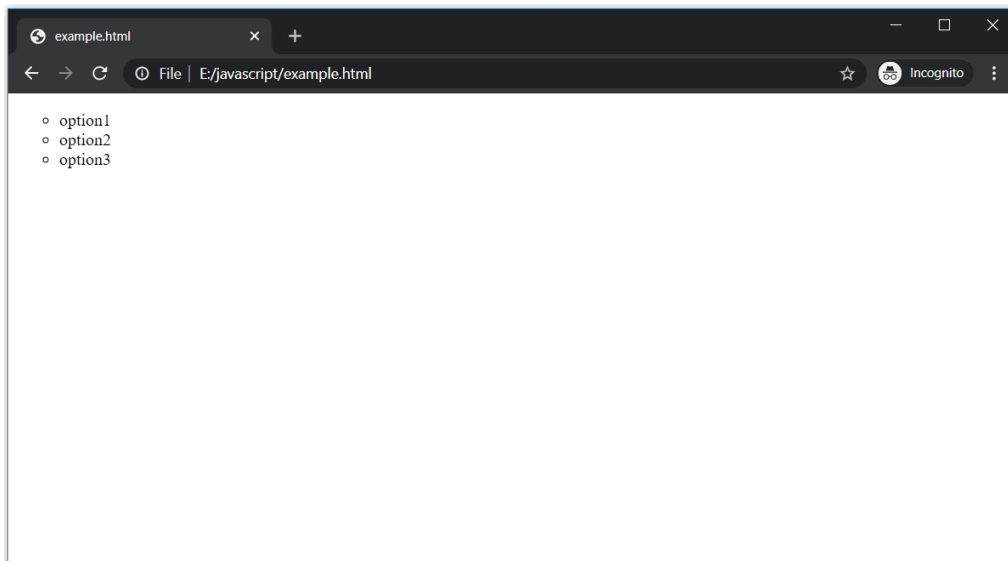
You can see, each option starts with a bullet.

Type attribute in the unordered list

The type attribute in the unordered list has four values; disc, circle, square, and none. By default, the type of unordered list is disc.

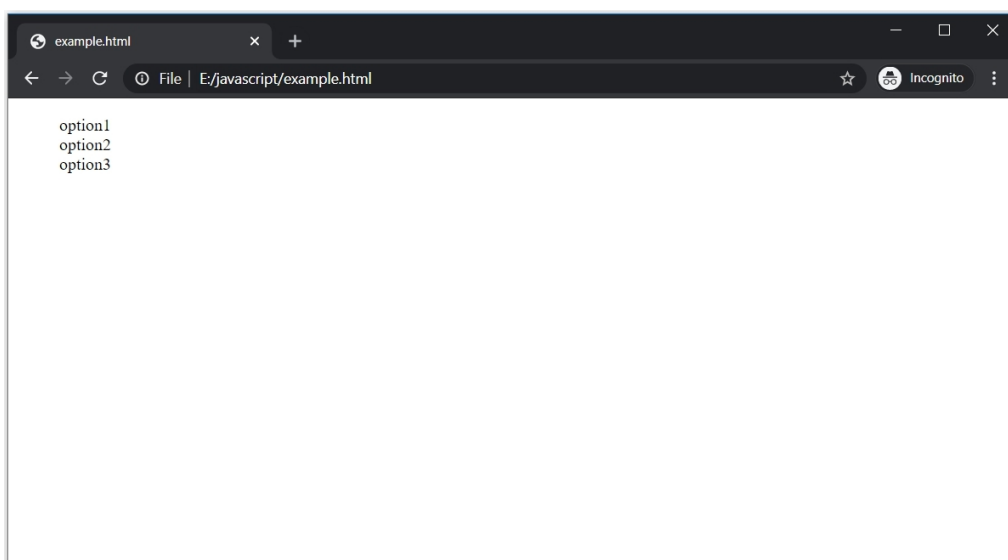
```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <ul type="circle">
6
7          <li> option1 </li>
8          <li> option2 </li>
9          <li> option3 </li>
10
11      </ul>
12
13 </body>
14
15 </html>
```

The type is "circle" for the unordered list.



If we do not want any symbol, we can use "none" as the type.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <ul type="none">
6
7          <li> option1 </li>
8          <li> option2 </li>
9          <li> option3 </li>
10
11      </ul>
12
13 </body>
14
15 </html>
```



Description list

The description list is used to create a list in which, **each option has a description with it.**

The `<dl>` tag defines a description list.

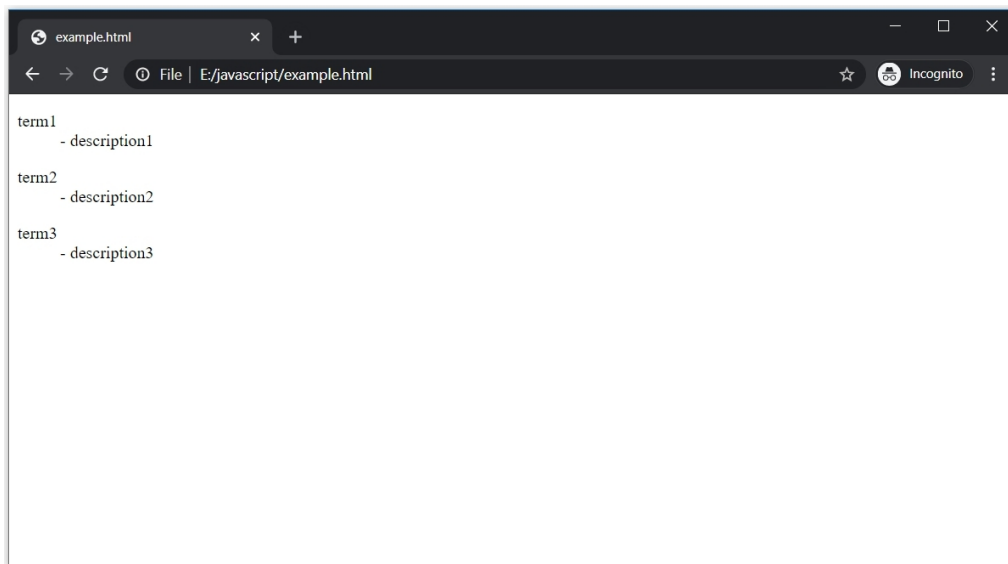
```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <dl>
6
7
8      </dl>
9
10 </body>
11
12 </html>
```

There are two additional tags - `<dt>` and `<dd>`.

The `<dt>` tag defines a term and the `<dd>` tag describes it.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <dl>
6          <dt> term1 </dt>
7          <dd>- description1</dd>
8      </dl>
9
10     <dl>
11         <dt> term2 </dt>
12         <dd>- description2</dd>
13     </dl>
14
15     <dl>
16         <dt> term3 </dt>
17         <dd>- description3</dd>
18     </dl>
19
20 </body>
21
22 </html>
```

There are three terms in the list, each having its own description.



Summary

- There are three types of lists in HTML - ordered, unordered, and description.
- The `<ol>` tag is used to create an ordered list.
- The `<ul>` tag is used to create an unordered list.
- The `<dl>` tag is used to create a description list.
- The `<li>` tag is used to create options in ordered and unordered lists.
- The `<dt>` tag defines a term in a description list while `<dd>` tag describes it.
- The type attribute in an ordered list can have 'a', 'A', 'i', and 'T' as its value. By default, it is a numerical list.
- The type attribute in an unordered list can have 'circle', 'square', and 'none' as its value. By default, it is a bulleted list.

Chapter 9: Forms

Forms are one of the most important parts of a web page because **they are used for user interaction**.

If we want information from a user, we need to create a form. There are various ways of getting input from the user, such as input fields, text areas, radio buttons, and checkboxes. They all come under the HTML forms.

<form> tag

The <form> tag defines a form in HTML. **Everything is written inside it.**

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      |
6      | <form>
7      | </form>
8      |
9      |
10 </body>
11
12 </html>
```

<input> tag

The <input> tag is the most important tag in HTML forms. This tag **defines how the user will provide the input. It all depends on the type attribute of the <input> tag.**

Type attribute

1. Input text field

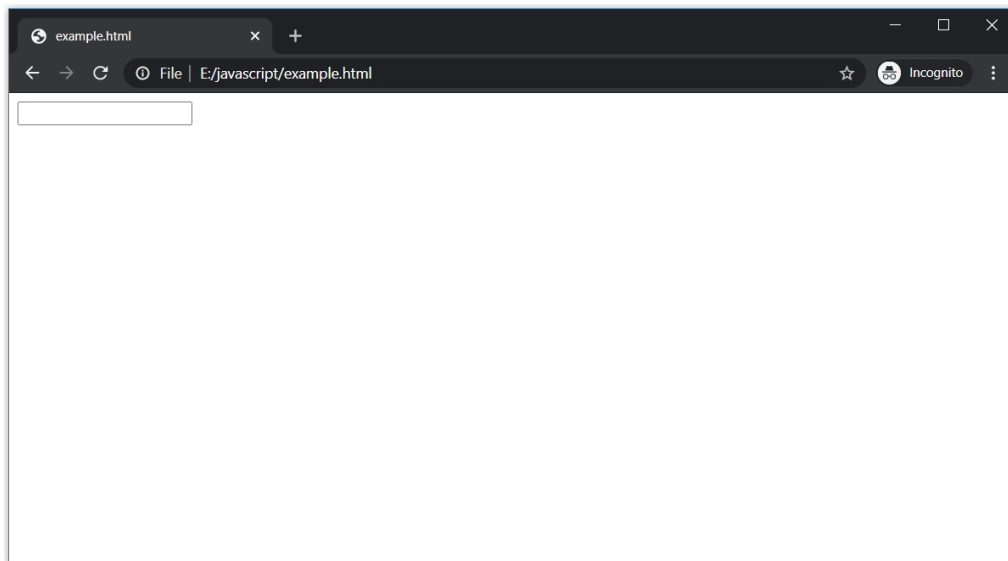
To create an input text field, set the value of the type attribute as "text".

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form>
7          <input type="text">
8      </form>
9
10 </body>
11
12 </html>

```

The `<input>` tag does not have a corresponding closing tag.



You can see the input field in the browser.

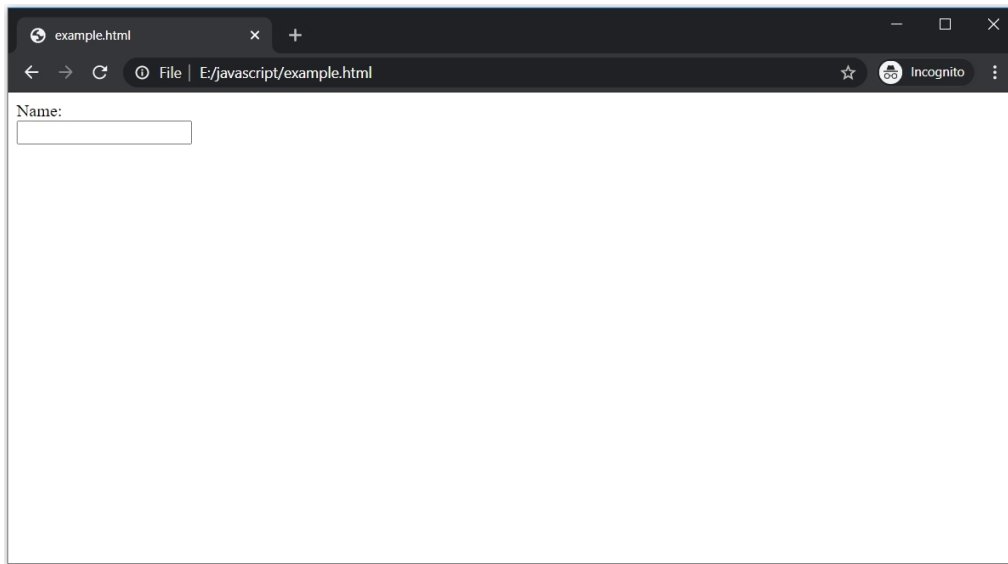
Let's make it more clear using the `<label>` tag.

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form>
7          <label>Name:</label> <br>
8          <input type="text">
9      </form>
10
11 </body>
12
13 </html>

```

The <label> tag has the information regarding the input. It can be used with any type of input.

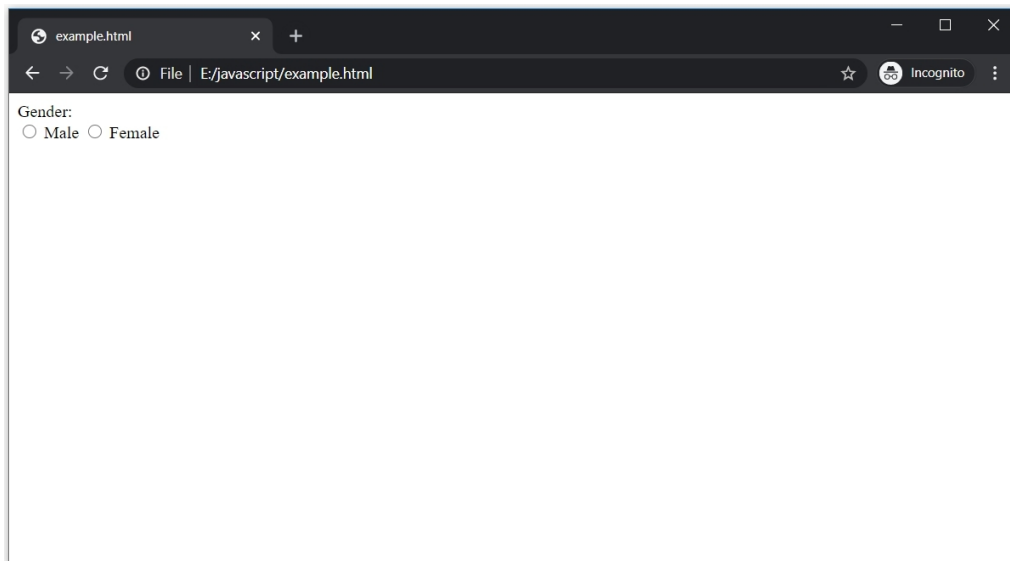


2. Radio buttons

If there are multiple choices and **the user can only select one** , use the radio buttons in such situations.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form>
7          <label> Gender: </label> <br>
8          <input type="radio">
9          <label> Male </label>
10         <input type="radio">
11         <label> Female </label>
12     </form>
13
14 </body>
15
16 </html>
```

There are two radio buttons.

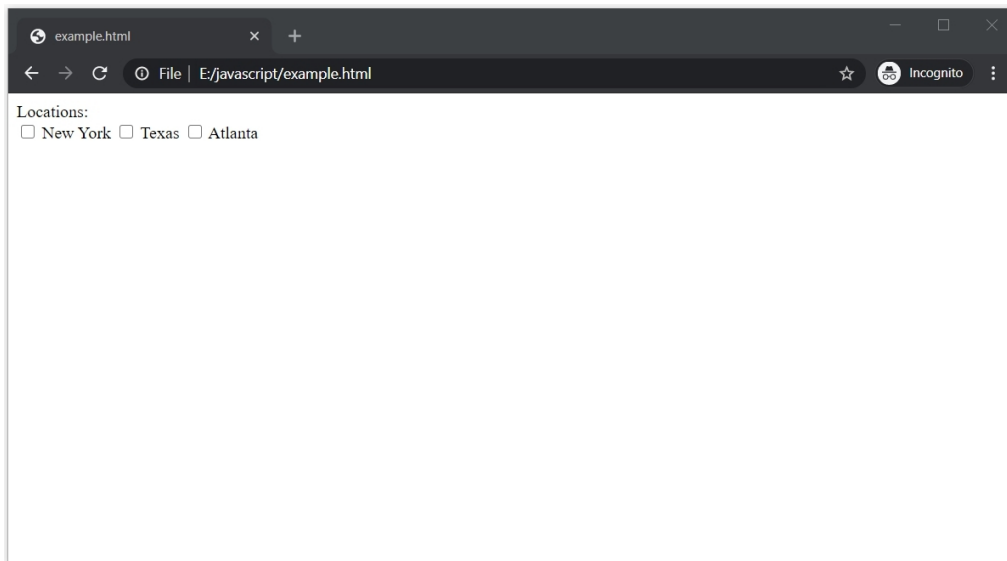


3. Checkboxes

Checkboxes are used where **a user can select multiple options.**

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form>
7          <label> Locations: </label> <br>
8          <input type="checkbox">
9          <label> New York </label>
10         <input type="checkbox">
11         <label> Texas </label>
12         <input type="checkbox">
13         <label> Atlanta </label>
14     </form>
15
16 </body>
17
18 </html>
```

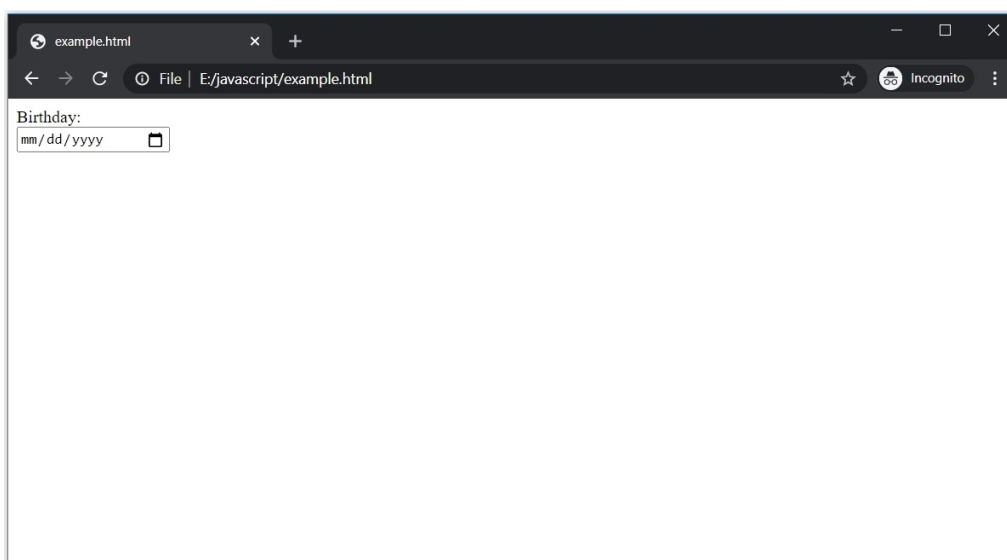
There are three checkboxes.



4. Date

To get a date as user input, set the value of the type attribute as date.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form>
7          <label> Birthday: </label> <br>
8          <input type="date">
9      </form>
10
11 </body>
12
13 </html>
```



5. Email

The email type is an input field that **only accepts when the input is a valid email**.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form>
7          <label> Email: </label> <br>
8          <input type="email">
9      </form>
10
11 </body>
12
13 </html>
```

There are several other input types such as:

- "file", allows user to upload a file
- "color", allows user to pick a color
- "number", text input that accepts only numbers.
- "month", allows the user to select a month and year.
- "time", allows the user to select the time.

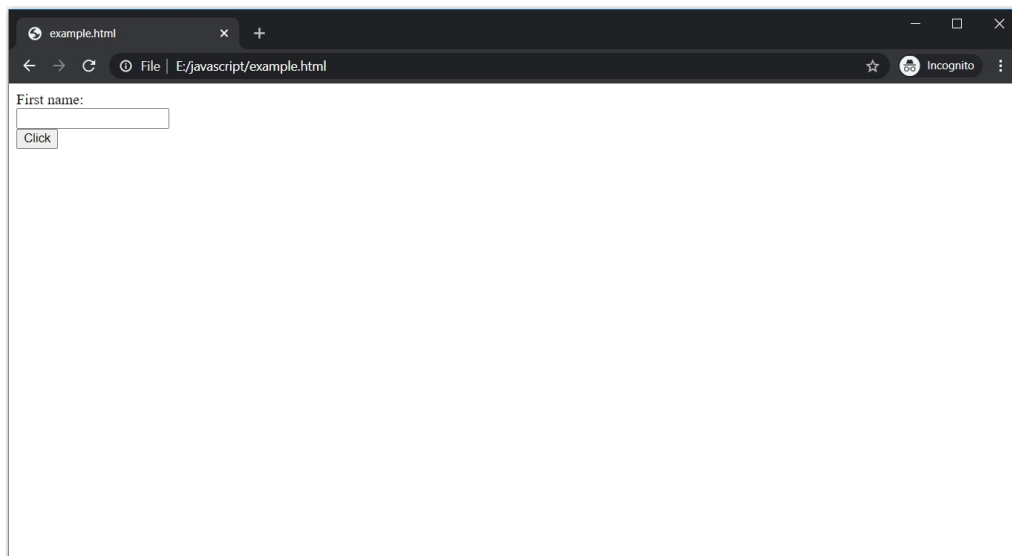
Button

A form is incomplete without buttons. There is always a button that submits the form data.

To create a button in a form, use "submit" as the type . It will create a submit form.


```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form>
7          <label> First name: </label> <br>
8          <input type="text"> <br>
9
10         <input type="submit" value="Click">
11     </form>
12
13 </body>
14
15 </html>
```

The value specified in the value attribute will appear on the button.



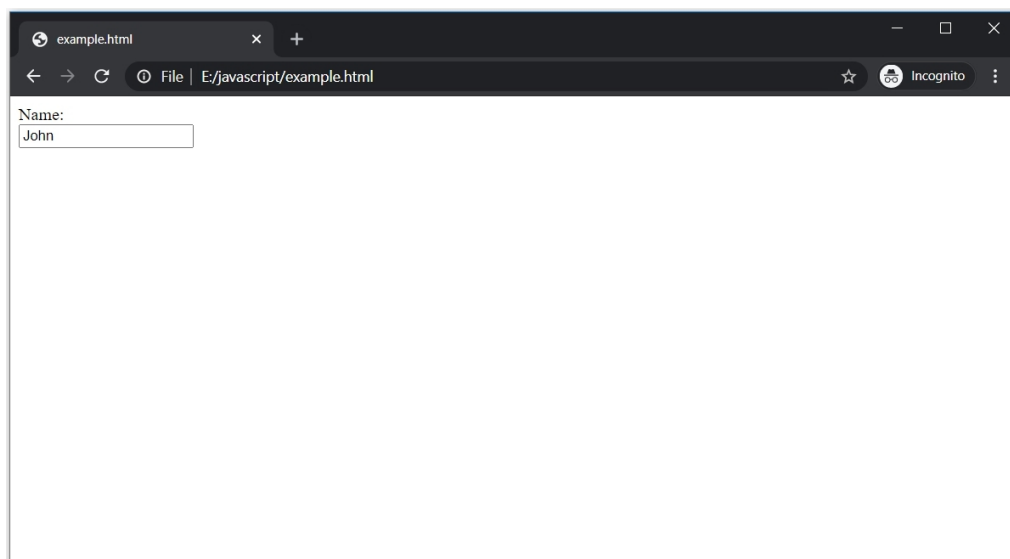
Important attributes

value attribute

The value attribute is used to give a **pre-defined value** to any input type.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form>
7          <label>Name:</label> <br>
8          <input type="text" value="John">
9      </form>
10
11 </body>
12
13 </html>
```

The value of the input text field is "John". Let's see in the browser.



You can see, the value appearing in the input field.

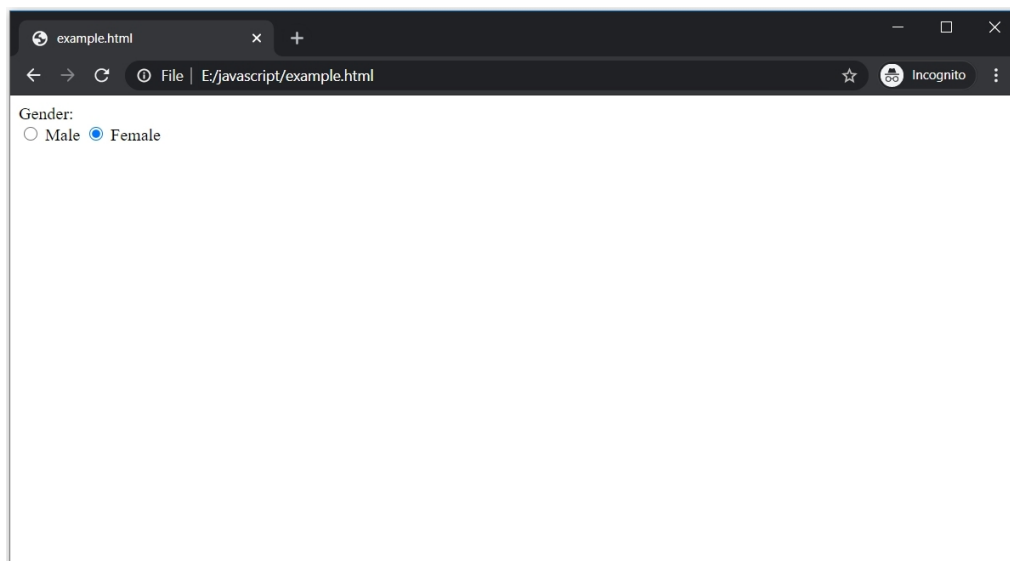
Similarly, the value can be set to any input type if it allows.

Checked attribute

The checked attribute is used with radio buttons and checkboxes. This attribute does not have any value. It is only placed in the input tag.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form>
7          <label> Gender: </label> <br>
8          <input type="radio">
9          <label> Male </label>
10         <input type="radio" checked>
11         <label> Female </label>
12     </form>
13
14 </body>
15
16 </html>
```

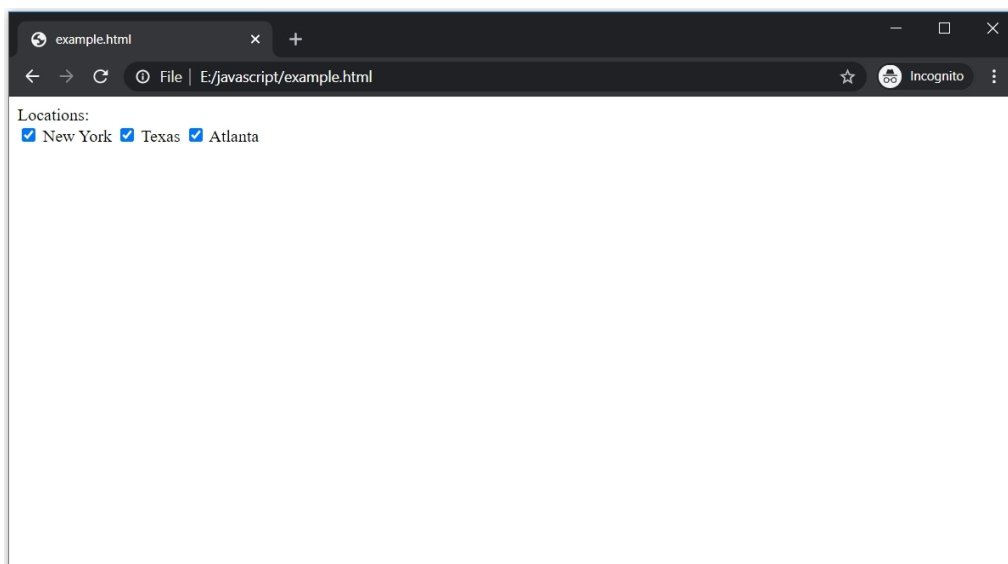
The second radio button is checked.



Similarly, the checked attribute can also be used with the checkboxes.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form>
7          <label> Locations: </label> <br>
8          <input type="checkbox" checked>
9          <label> New York </label>
10         <input type="checkbox" checked>
11         <label> Texas </label>
12         <input type="checkbox" checked>
13         <label> Atlanta </label>
14     </form>
15
16 </body>
17
18 </html>
```

All the checkboxes are checked.

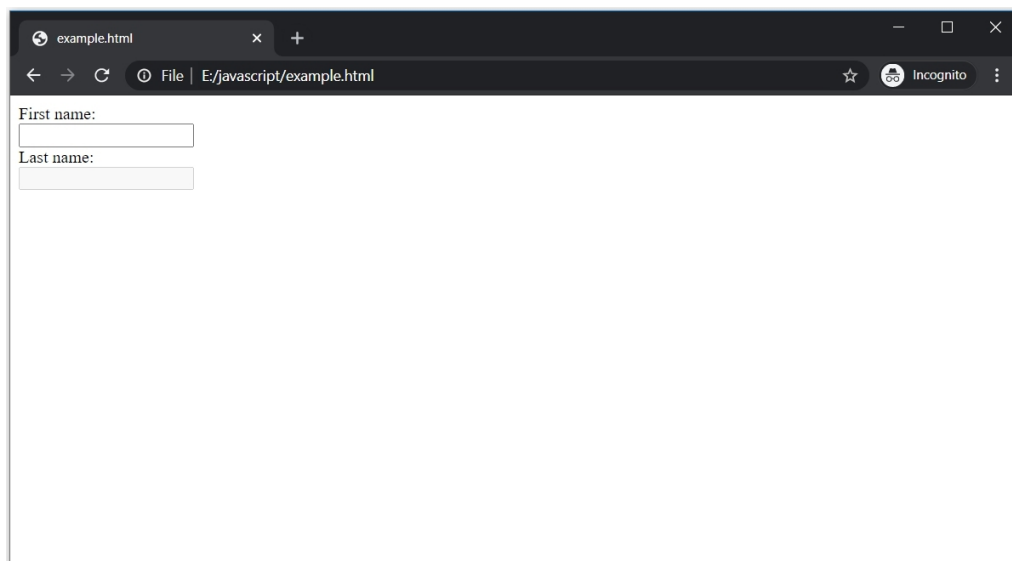


disabled attribute

To disable an input field, use the disabled attribute. By disable, it means, **user is not allowed to use the input field.**

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form>
7          <label> First name: </label> <br>
8          <input type="text"> <br>
9
10         <label> Last name: </label> <br>
11         <input type="text" disabled>
12     </form>
13
14 </body>
15
16 </html>
```

The second input text field is disabled.



required attribute

Any input field with the required attribute cannot be skipped. **It is mandatory to provide value to that field which is required.**

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form>
7          <label> First name: </label> <br>
8          <input type="text"> <br>
9
10         <label> Last name: </label> <br>
11         <input type="text" required>
12     </form>
13
14 </body>
15
16 </html>

```

placeholder attribute

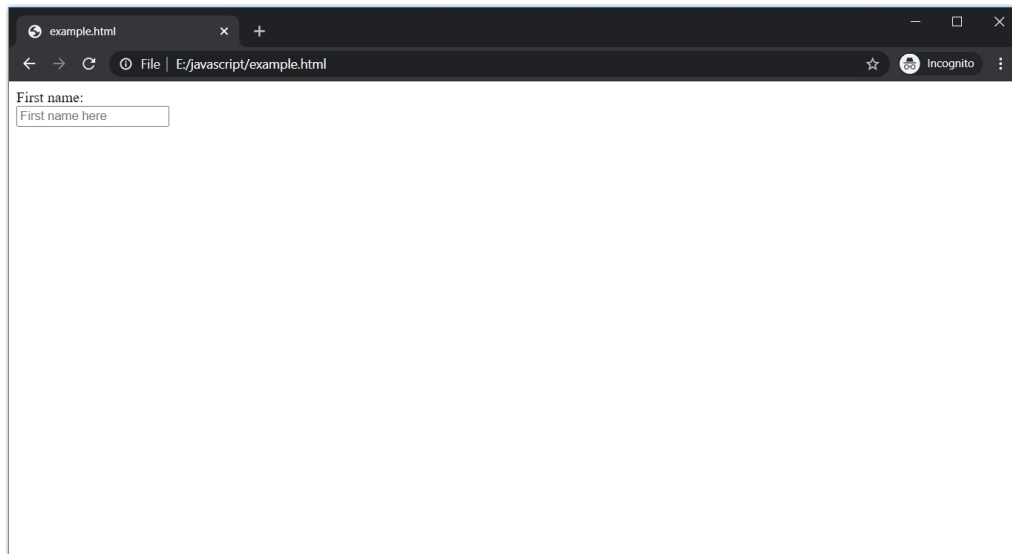
A placeholder is a text that appears in an input field to indicate the expectations. The text is just for display, it is not the value of the field.

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form>
7          <label> First name: </label> <br>
8          <input type="text" placeholder="First name here"> <br>
9
10     </form>
11
12 </body>
13
14 </html>

```

The value of the placeholder will appear in the input text field.



Form attributes

We discuss some of the important and commonly used input attributes. Now, let's discuss some of the important attributes of the `<form>` tag.

action attribute

The action attribute in a form defines **which URL will invoke when the form is submitted**. The input values are delivered to the specified URL.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form action="/action_.php">
7
8      </form>
9
10 </body>
11
12 </html>
```

The value of the action attribute should be a URL.

method attribute

It is also necessary to specify the HTTP method when a form is submitted. The method attribute is used to specify the method. It can be either POST or GET.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form action="/action_.php" method="POST">
7
8      </form>
9
10 </body>
11
12 </html>
```

novalidate attribute

The `<input>` tag has few attributes for validation, such as the `required` attribute. The form will not submit until everything in the form is validated. But we can also ignore the validations by using the `novalidate` attribute.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <form action="/action_.php" method="POST" novalidate>
7
8      </form>
9
10 </body>
11
12 </html>
```

`novalidate` is a boolean attribute so it does not requires any value.

Summary

- Forms are used to get input from the users.
- The `<form>` tag is used to define a form and the `<input>` tag is used to define the way in which the user will provide input.
- There are several types of input in HTML. The `<input>` tag has the `type` attribute that defines the input field. The types include text, radio, checkbox, and many more.
- To create a button, specify the type of `<input>` tag as "submit". It will create a submit button for the form.
- The `value` attribute is used to give value to the input field.
- There are several other attributes such as `checked`, `disabled`, `placeholder`, etc which are useful for presentation and validation.

- The action attribute of the <form> tag describes the action to be taken when the form is submitted while the method attribute describes the method of the action.

Chapter 10: Media

Multimedia has become a trend in modern websites. We may find videos and audios playing automatically when we visit a particular website, or we may find options to play the video or audio. HTML provides separate tags for video and audio with some useful attributes.

<video> tag

As the name suggests, **the <video> tag is used to display videos in the browser.**

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      |
6      | <video>
7      |
8      | </video>
9      |
10 </body>
11
12 </html>
```

<source> tag

The <source> tag is used to define the source of the video. It has an src attribute that holds the URL of the video.

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      |
6      | <video>
7      | | <source src="videoplayback.mp4">
8      | | </source>
9      | </video>
10 </body>
11
12 </html>
```

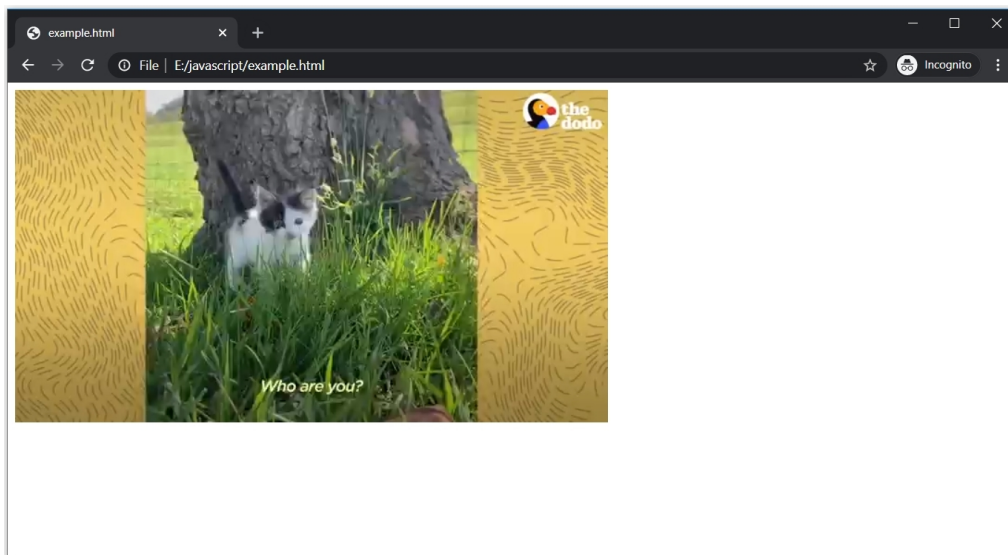
We also need to mention the type of video using the type attribute.

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <video>
7          <source src="videoplayback.mp4" type="video/mp4">
8      </video>
9
10 </body>
11
12 </html>

```

Observe how the video looks in the browser.



The video appears on the screen it will not play because there are not options for play or pause.

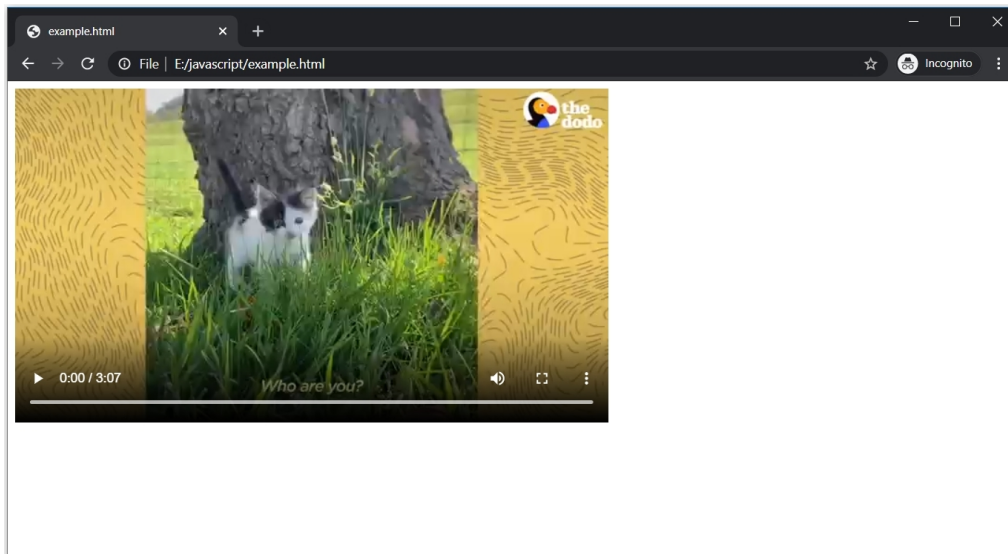
controls attribute

The `<video>` tag has a boolean attribute called `controls`. **This attribute provides controlling options for the video.**

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <video controls>
7          <source src="videoplayback.mp4" type="video/mp4">
8      </video>
9
10 </body>
11
12 </html>

```



You can see, the options such as play/pause button and length of the video are available now.

autoplay attribute

The autoplay attribute is another Boolean attribute of the `<video>` tag. When it is specified, **the video starts playing automatically as soon as the page loads.**

```
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6      <video controls autoplay>
7          <source src="videoplayback.mp4" type="video/mp4">
8      </video>
9
10 </body>
11
12 </html>
```

`<audio>` tag

The `<audio>` tag is used to define audio on a web page.

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6  <audio>
7
8  </audio>
9
10 </body>
11
12 </html>

```

Similar to the <video> tag, the <source> tag is also used here.

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6  <audio>
7  | <source src="videoplayback.mp3" type="audio/mpeg">
8  </audio>
9
10 </body>
11
12 </html>

```

Moreover, the <audio> tag also has the controls and autoplay attributes.

```

1  <!DOCTYPE html>
2  <html>
3
4  <body>
5
6  <audio controls autoplay>
7  | <source src="videoplayback.mp3" type="audio/mpeg">
8  </audio>
9
10 </body>
11
12 </html>

```

Summary

- The <video> and <audio> tags are used to define videos and audios in HTML, respectively.
- The <source> tag is to define which video or audio is to be played. It is also necessary to define the type using the type attribute in the <source> tag.